

Flying Ball in a Hoop

**Design of new didactic devices
for teaching of control engineering**

Master Thesis – Academic Year 2025-2026

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Context

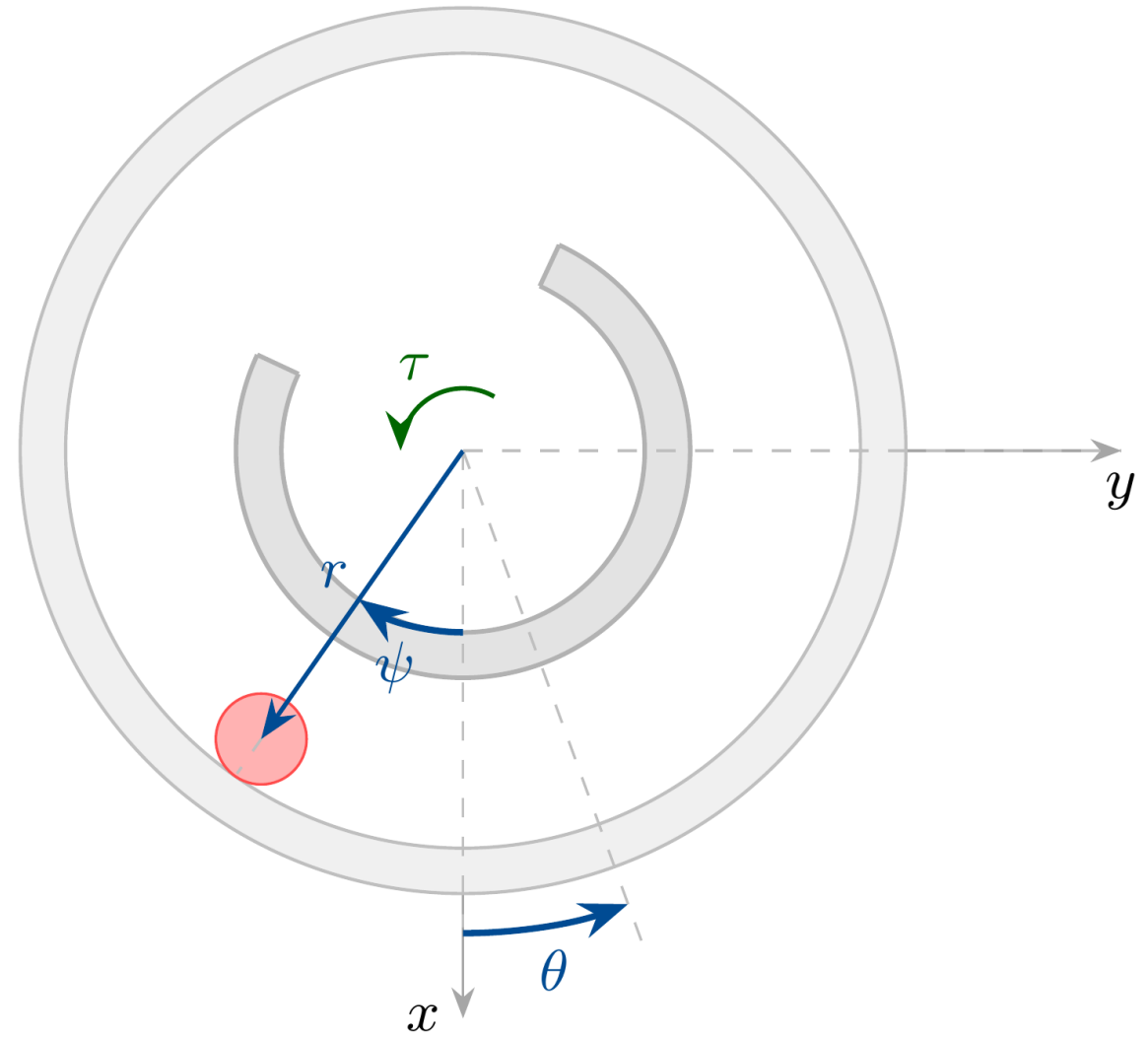
Why the ball-and-hoop?

System configurations

- Single-hoop: 2 DOFs (ψ, θ), 1 actuator (τ)
→ Underactuation
- Double-hoop: hybrid modes

Why is it worth studying?

- **Classic benchmark family** – inverted pendulum, ball-and-beam (*Horáček, 2000*)
- **Industrial analogue** – Liquid slosh in moving tanks (*Wellstead, 1983*)
- **Intuitive and observable** – Ball and hoops position directly visible
- **Extensible** – Double-hoop, growing task complexity (*Gurtner et Zemánek, 2017*)



Double hoop configuration : starting point



Context

Starting point and contribution

Starting point

Double-hoop benchmark, introduced by *Gurtner & Zemánek (2017)*.

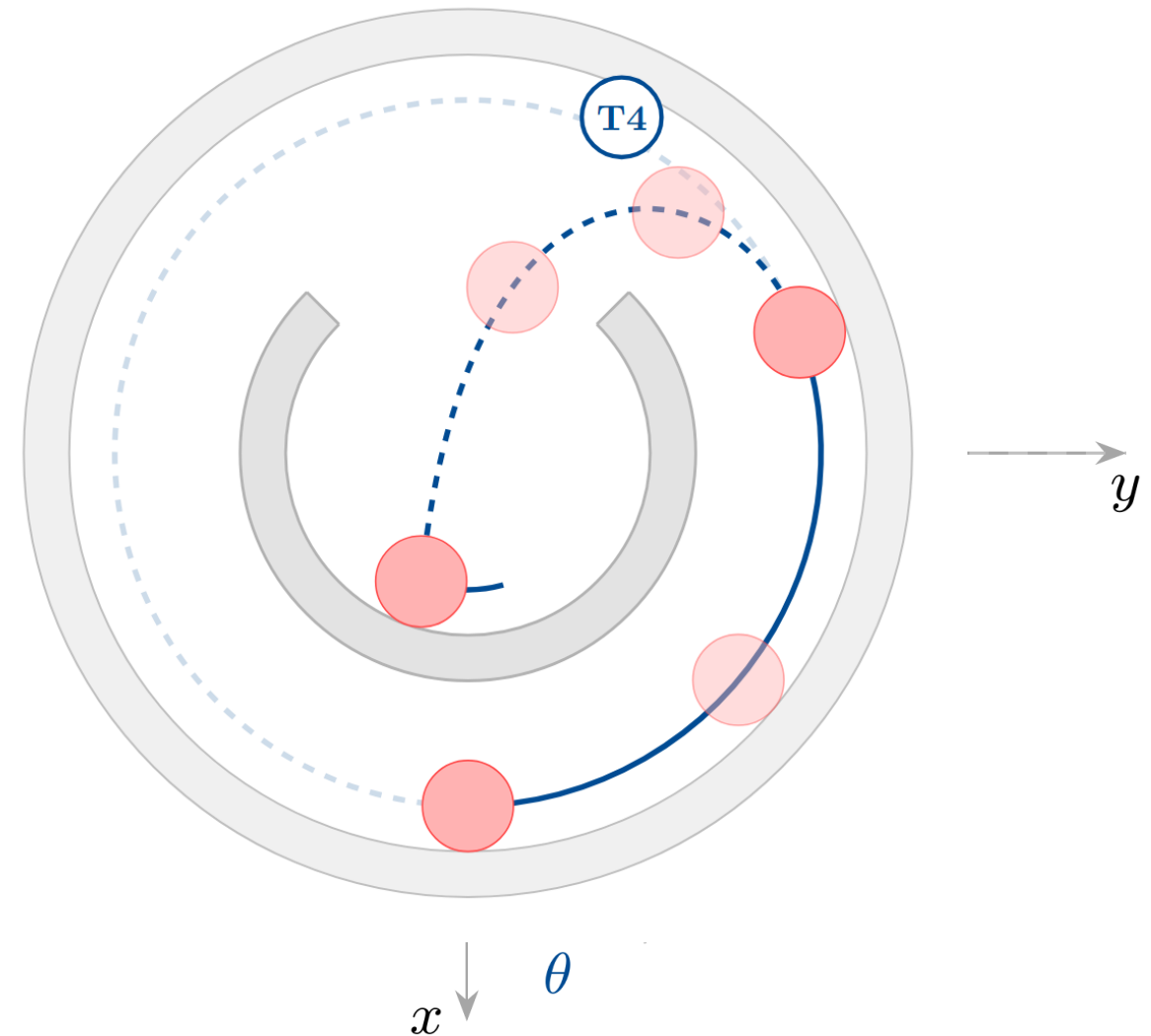
Goal: Build and control a ball-and-hoop system

Approach

- Mathematical Modelling
- Prototype building and validation
- Control tasks

Control Tasks

- T1 – Stabilisation
- T2 – Tracking around 0°
- T3 – Looping
- T4 – Flying Ball



What is needed to create a ball-and-hoop system?

Mathematical Modelling

Hybrid formulation

Modes Set of differential equations

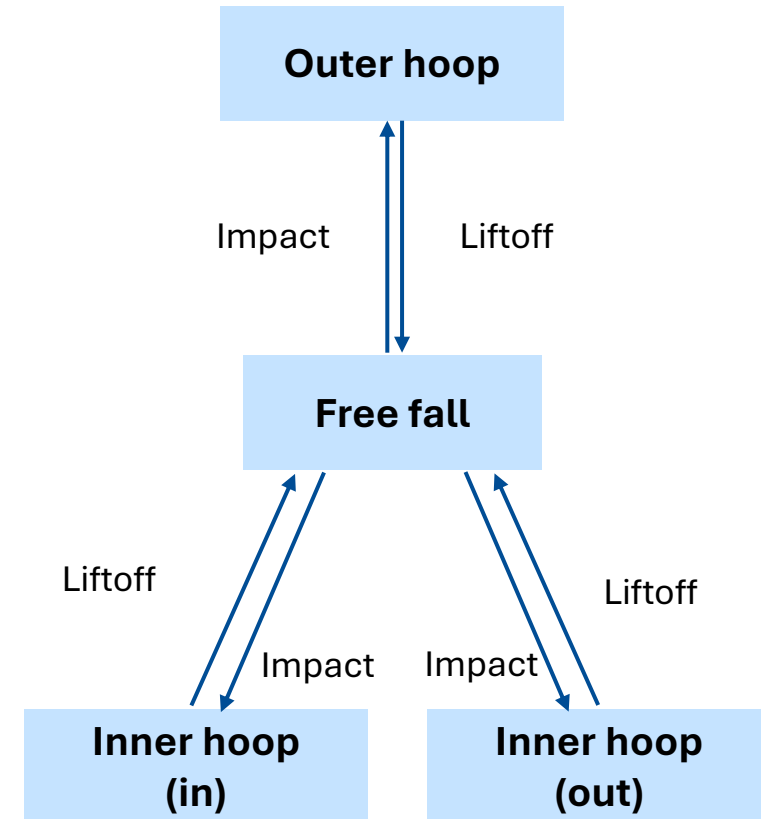
- 3 rolling modes
- 1 free fall mode

Guards Condition triggering a transition when they vanish

- Liftoff
- Impact

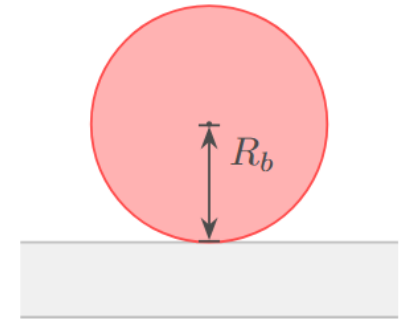
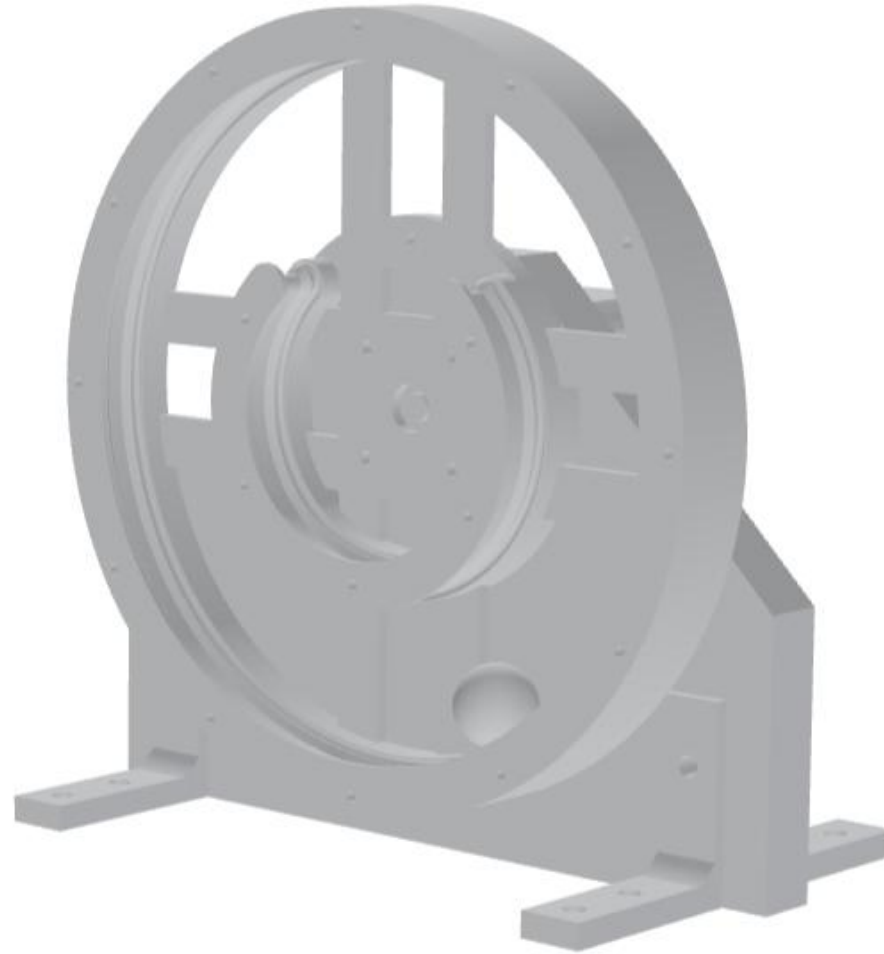
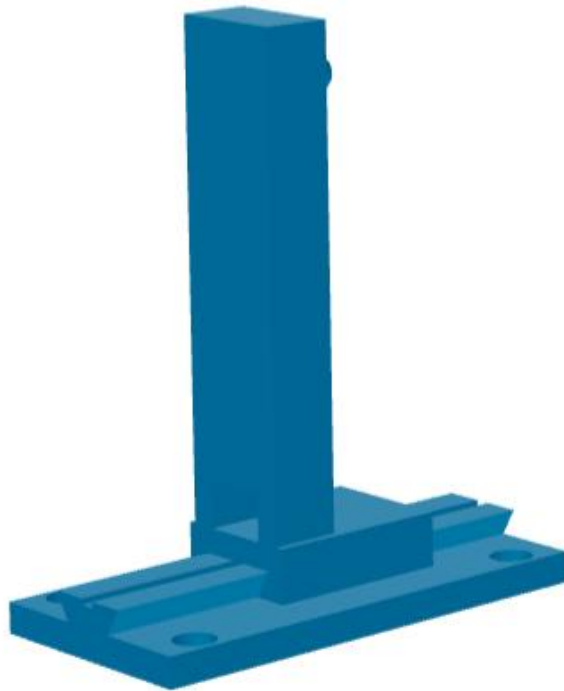
Reset Map How the state changes at a transition

- Continuous at liftoffs
- Discontinuous at impacts

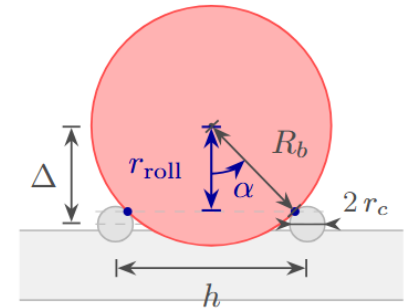


Numerically verified on a MATLAB simulation: modes, transitions, guards, energy conservation

Prototype Construction

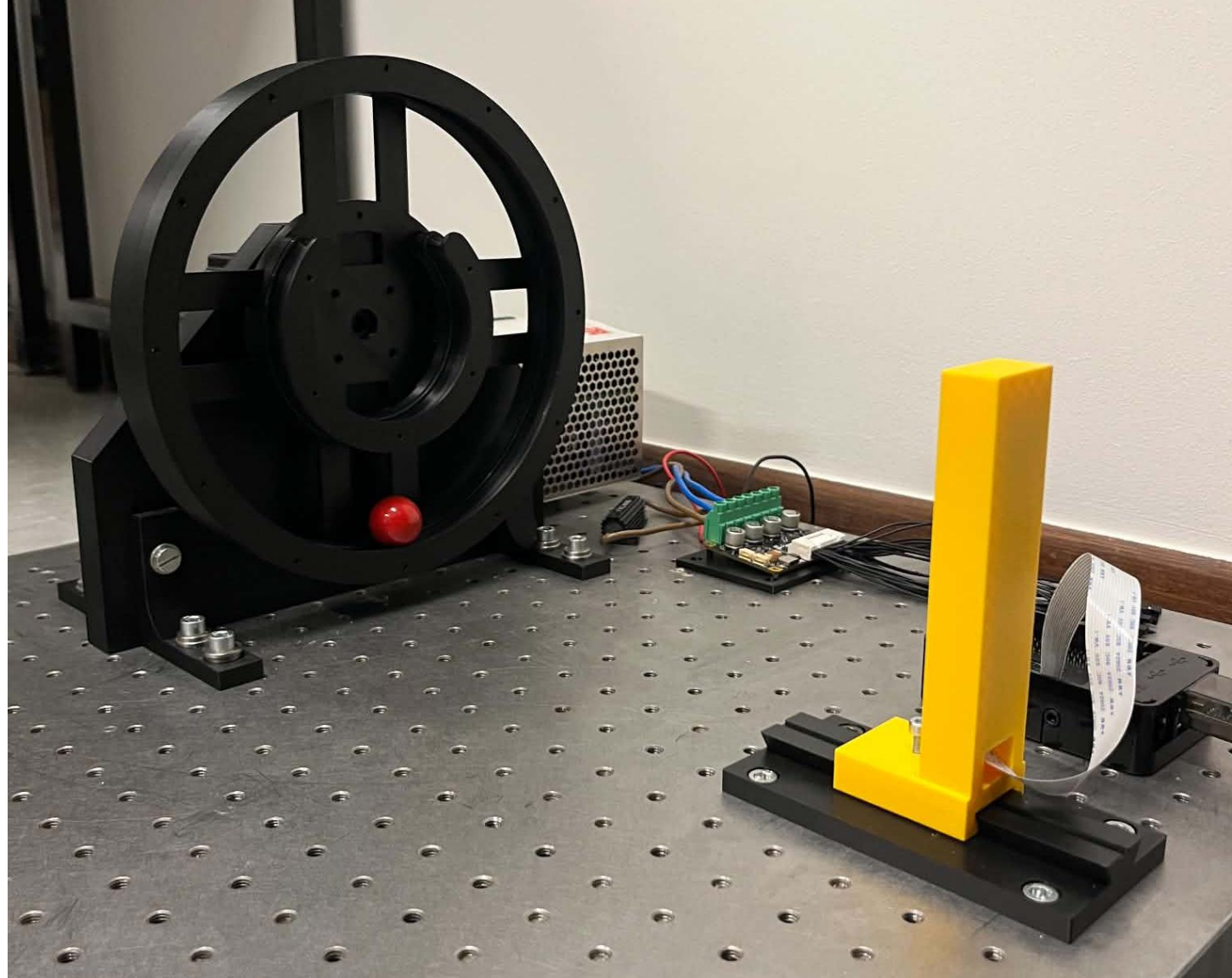


Ideal contact case



Rails contact case

Prototype Construction



The bench is now built, but still has to be validated

Prototype

Parameters identification and validation



Identification

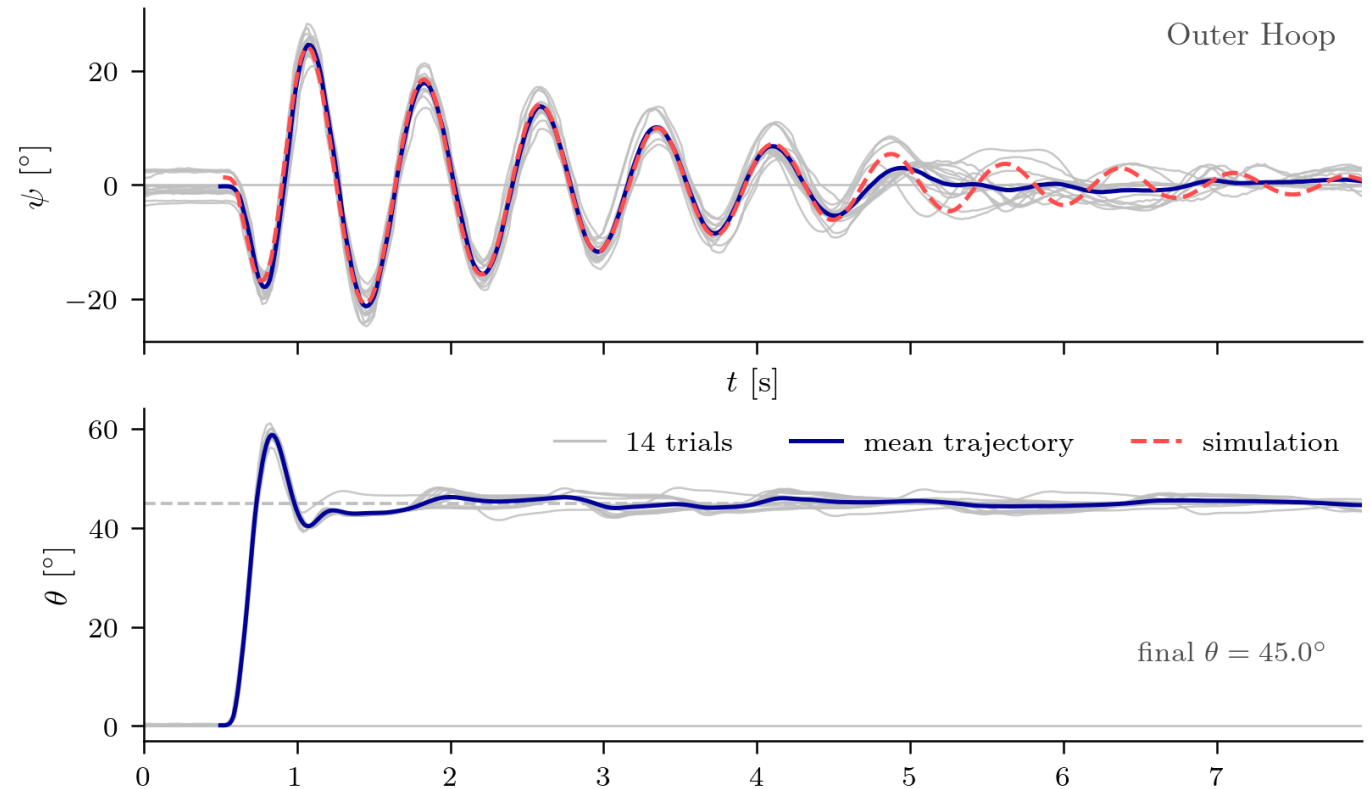
- Hoops and motor inertia
- Hoops and motor friction
- Ball friction
- ➔ Validation

Protocol

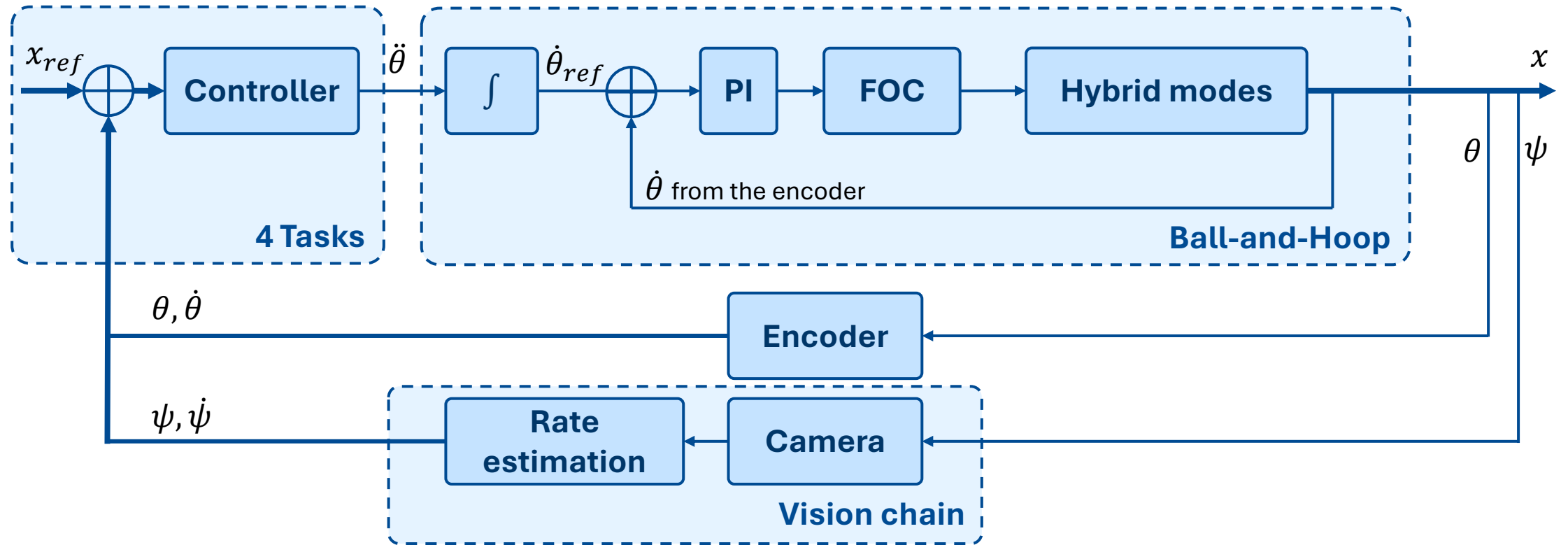
- Simulation confronted with bench
- Step response (Step of $\theta = 45^\circ$)

Results

- Mean RMSE: $RMSE = 2.47^\circ$
- Freq. Error: $e(f_0) = 3.46\%$

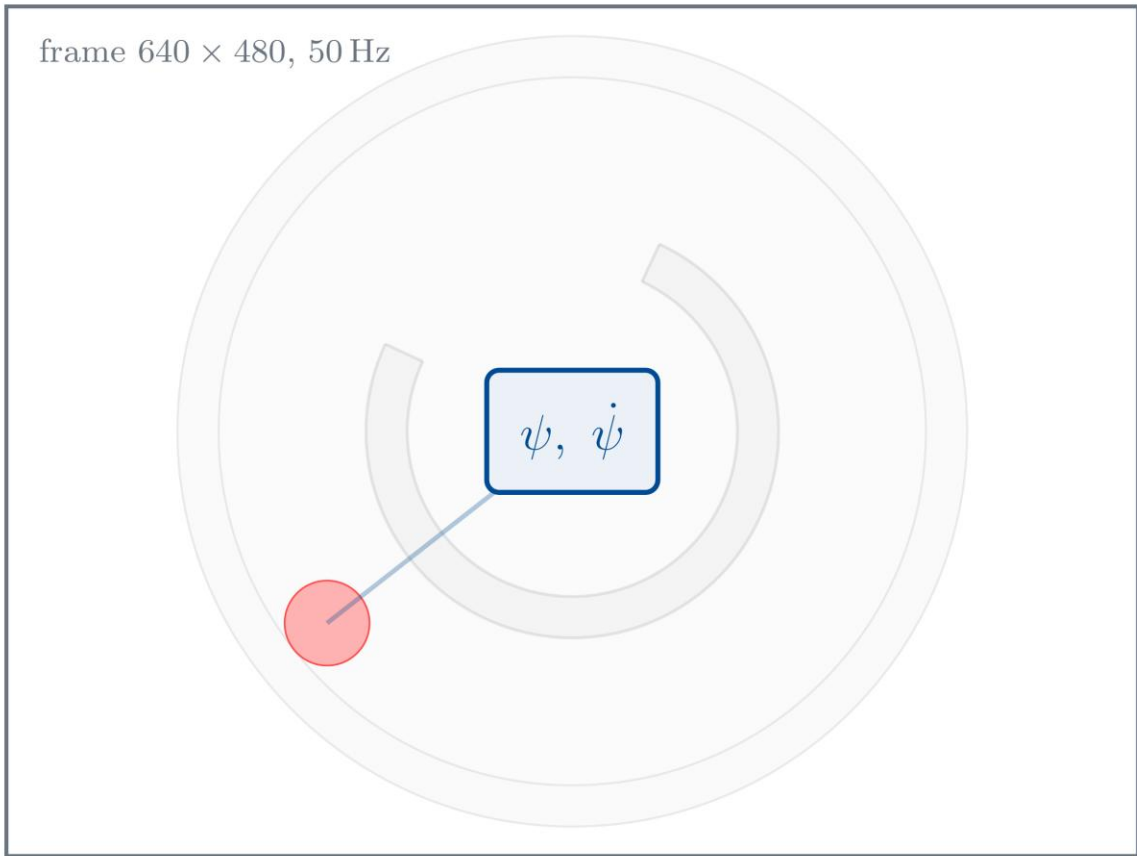
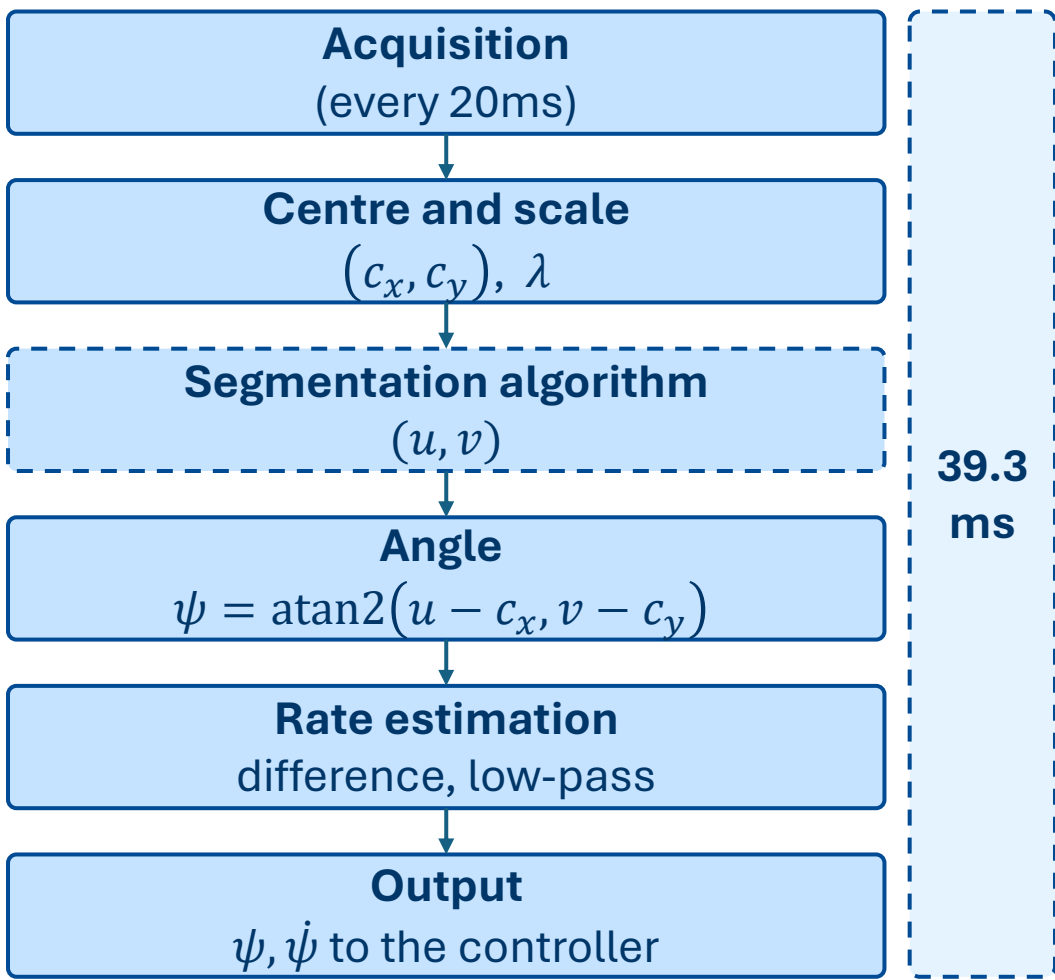


The model is validated on its continuous parts



Prototype

Vision Chain



Linear Control Design



Open-Loop margins

LQR controller

- Full state feedback

$$u(t) = x_{ref}(t) - K x(t)$$

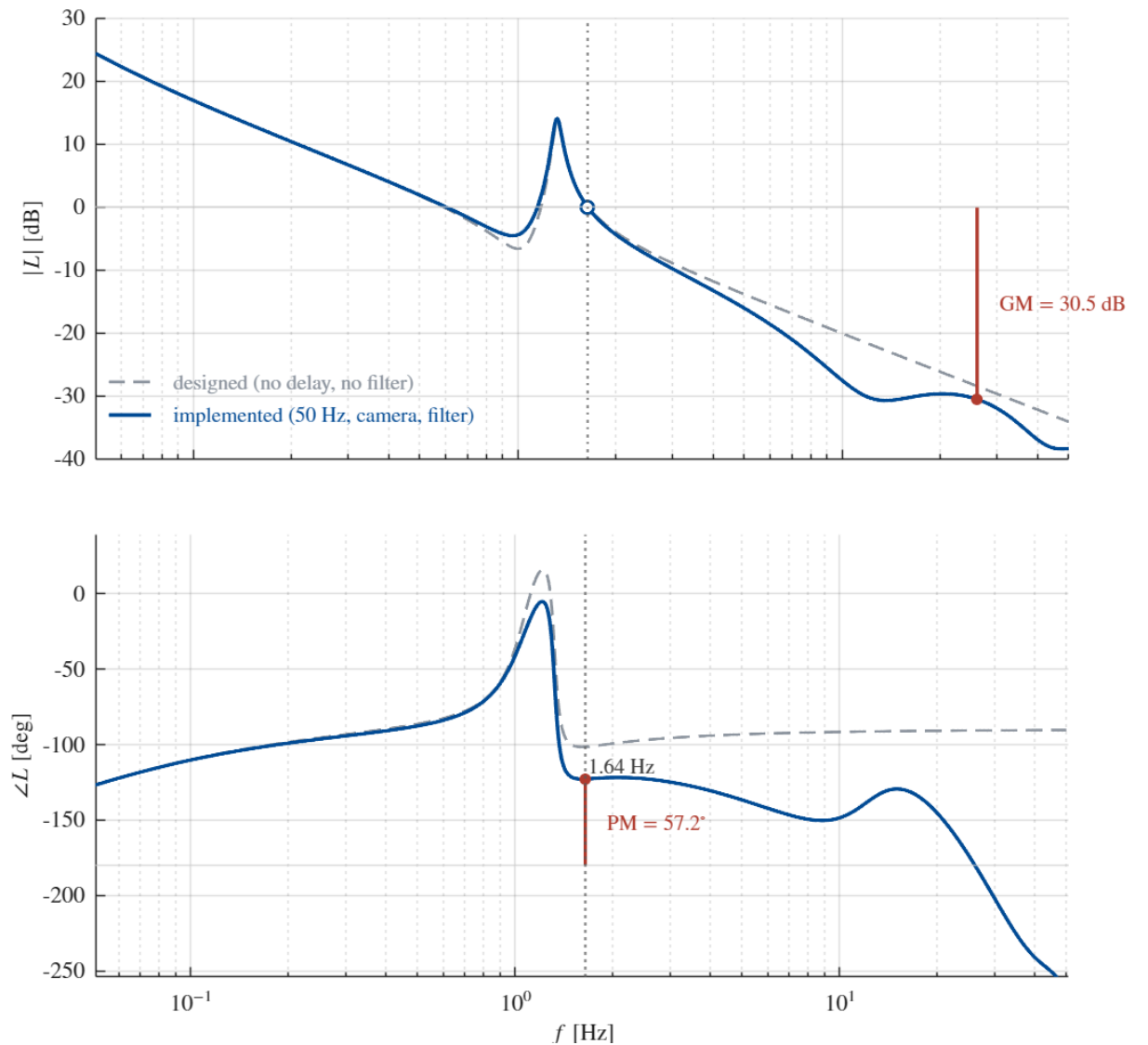
- Extended to TVLQR later

Resulting loop

- LQR
- Camera delay
- Zero-Order-Hold (ZOH)
- Low-pass filter

Resulting loop

- **GM:** 30.5 dB
- **PM:** 57.2°

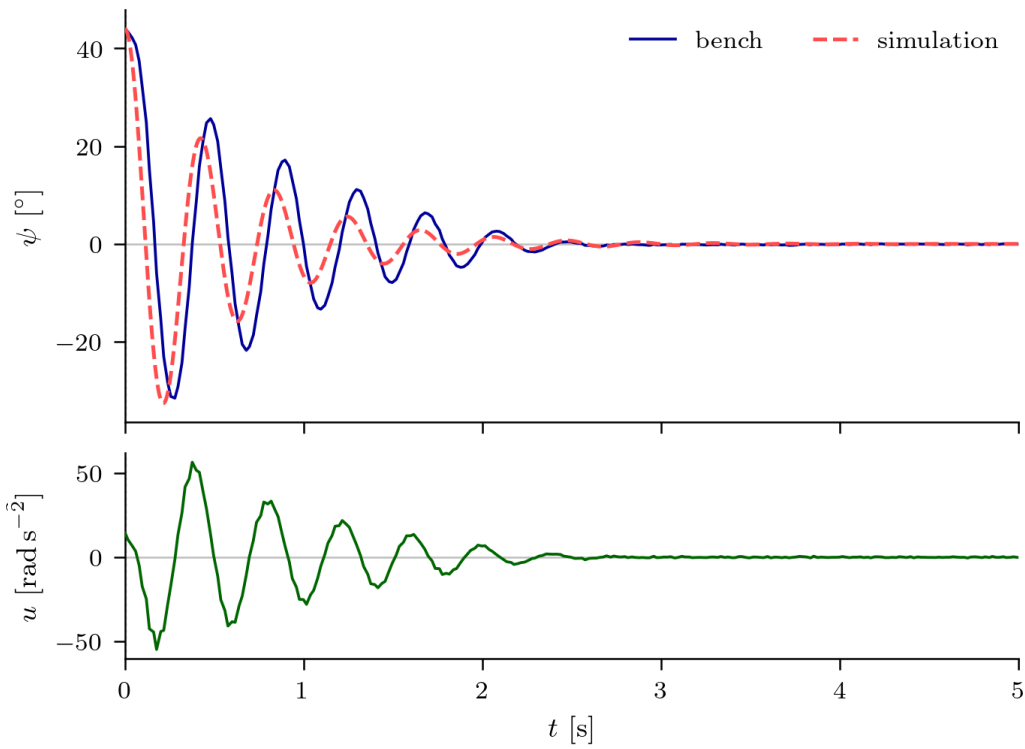




Linear Control Design

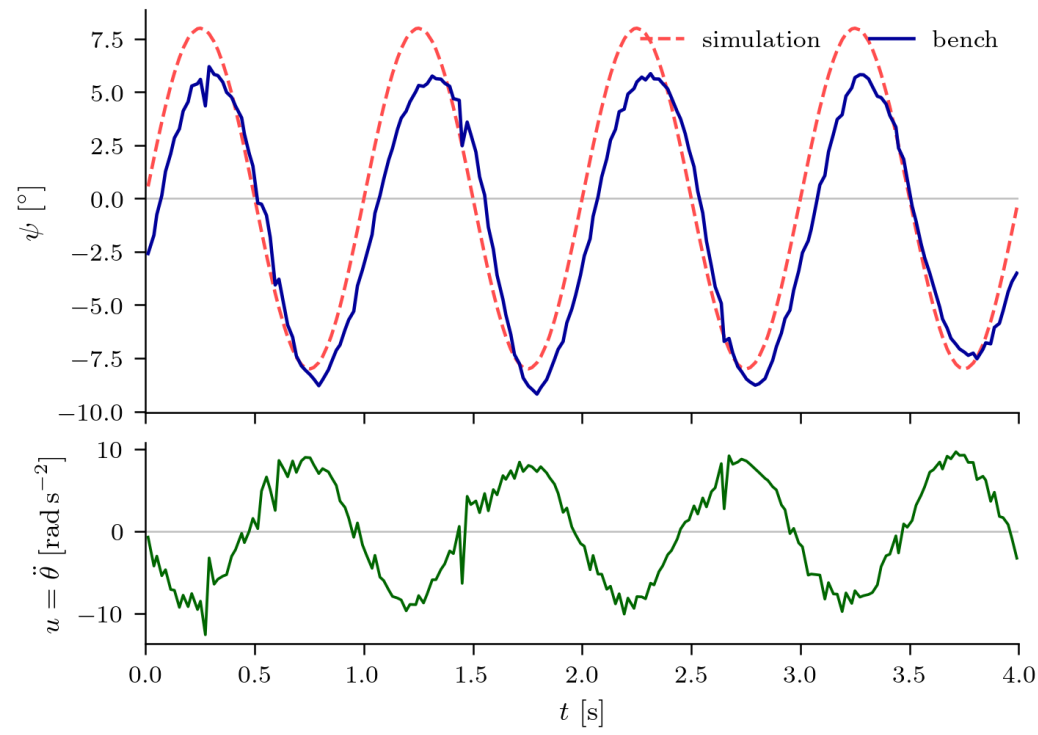
T1 and T2 results

T1 – Stabilisation



Recovers well beyond the linearisation region

T2 – Tracking around 0°



The ball is tracked with a slight gain reduction



Trajectory Generation

Leave the linear zone

T3 – Changes

- T1, T2 : linearisation around equilibrium
- T3 : full revolution → Linear methods don't work

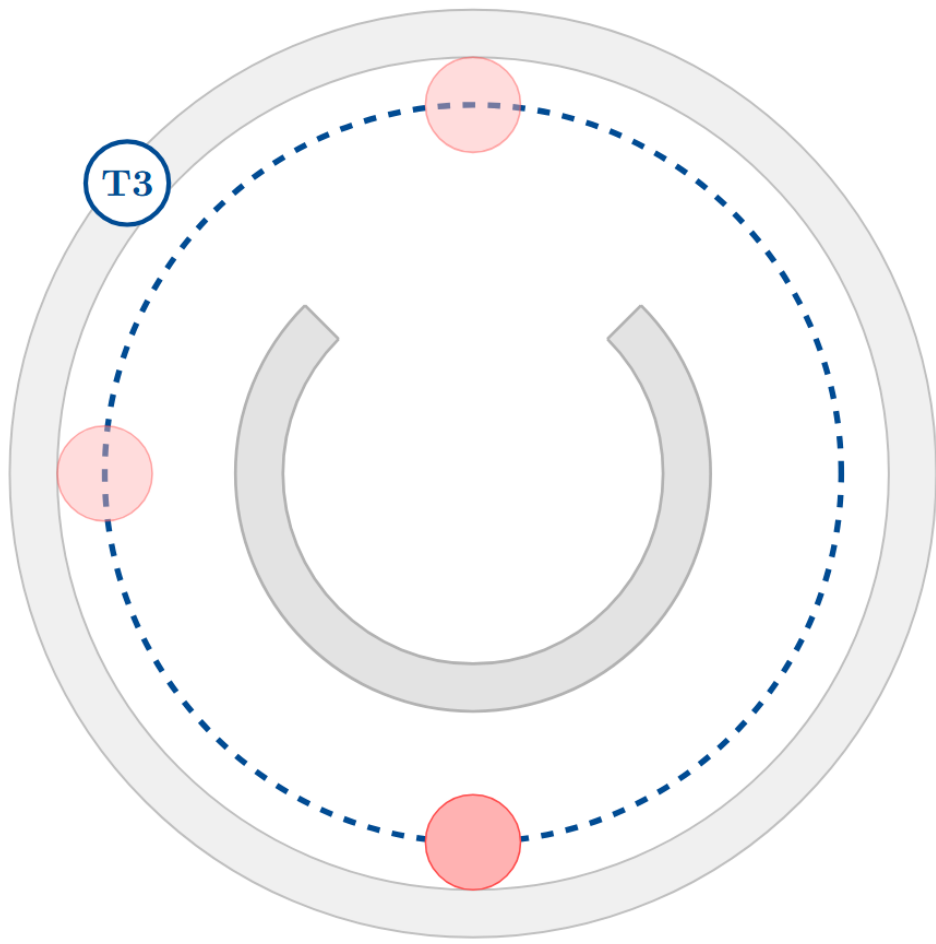
Problem formulation

State $x = (\theta, \dot{\theta}, \psi, \dot{\psi})$, **input** $u = \ddot{\theta}$

$$\min_{x(\cdot), u(\cdot), T_f} J = \int_0^{T_f} u(t)^2 dt$$

Under constraints

- Dynamics: $\dot{x} = f(x, u)$, outer hoop rolling
- Saturation: $|u| \leq u_{max}$
- Boundaries: $x(0) = 0, \quad \psi(T_f) = -2\pi$
- Contact: $N(x, u) \geq 0$



Generate a nominal trajectory (direct collocation) and stabilise the system around it (TVLQR)



Trajectory Generation

Direct Collocation

Trajectory Generation

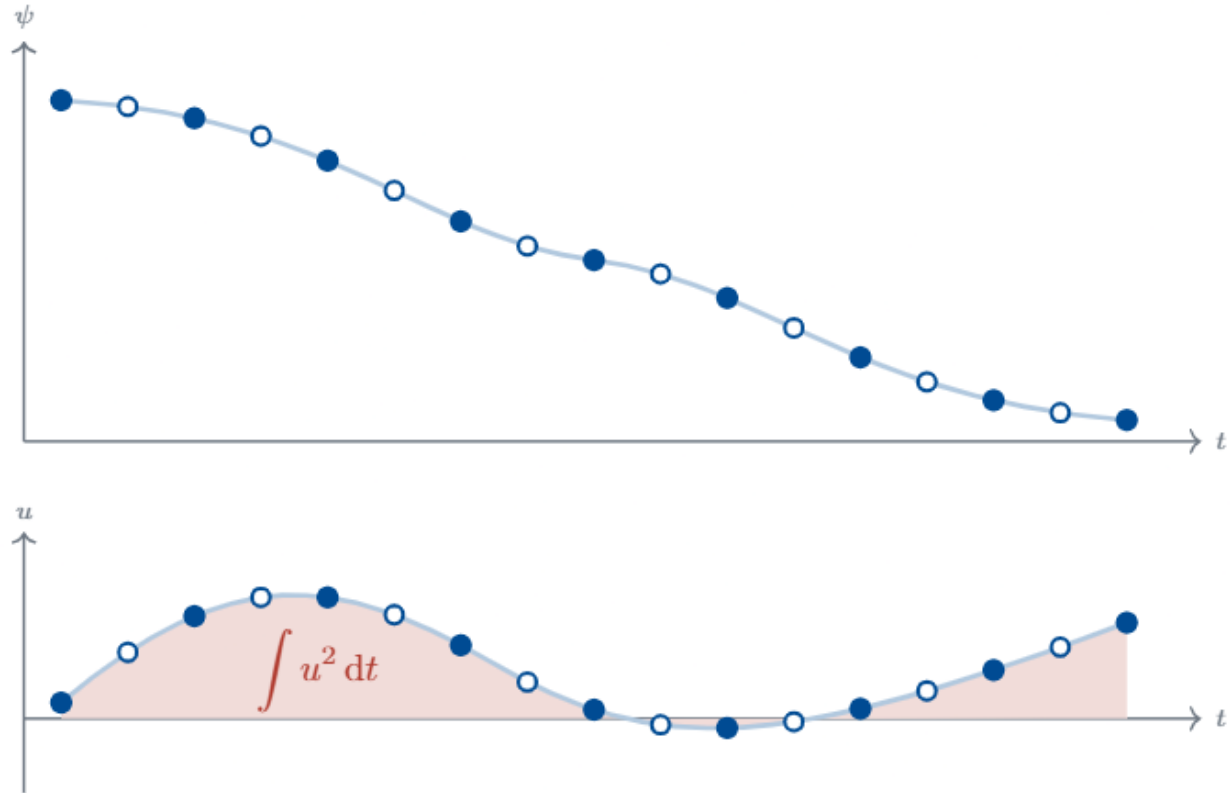
- Start from initial guess
- Discretisation in K nodes, separated by h_k
- Defects calculation (Hermite-Simpson)

$$x_{c,k} = \frac{x_k + x_{k+1}}{2} + \frac{h_k}{8} (f_k - f_{k+1})$$

$$x_{k+1} - x_k = \frac{h_k}{6} (f_k + 4f_{c,k} + f_{k+1})$$

- Constraints imposed at nodes and midnodes
- Resolution: CasADi

$$J = \int_0^{T_f} u(t)^2 dt$$



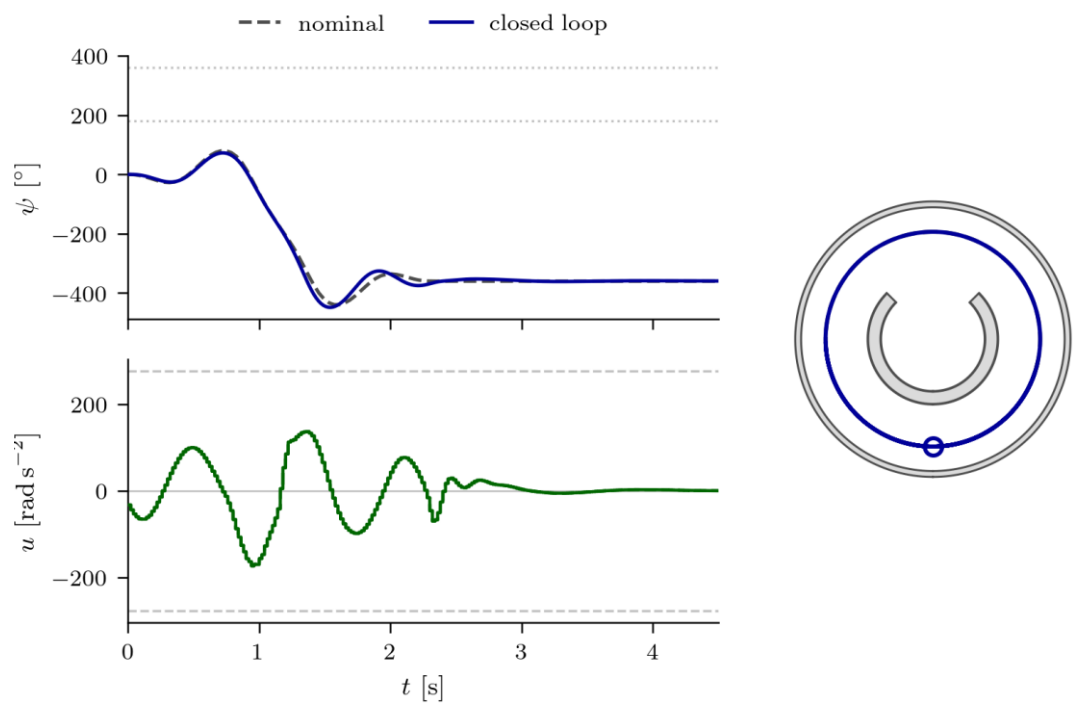
Feedback using TVLQR: $u(t) = u^*(t) - K(t) \delta x(t)$

Looping Control Design

Results

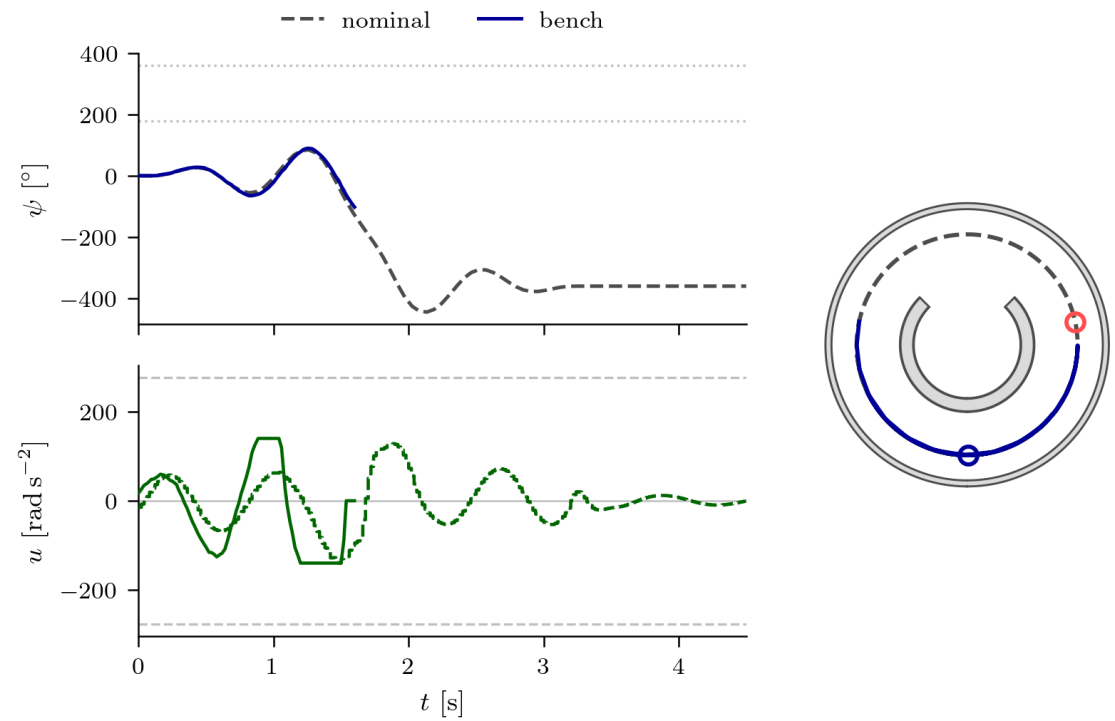


Simulation



All constraints are satisfied

Bench



The ball leaves the track before the top
At about the same angle in every trial



Looping Control Design

Where does the failure come from?

Actuator saturation?

Below saturation value.

Camera tracking loss ?

High velocities appear before failure.

Excentricity error?

Several θ_0 lead to the same failure.

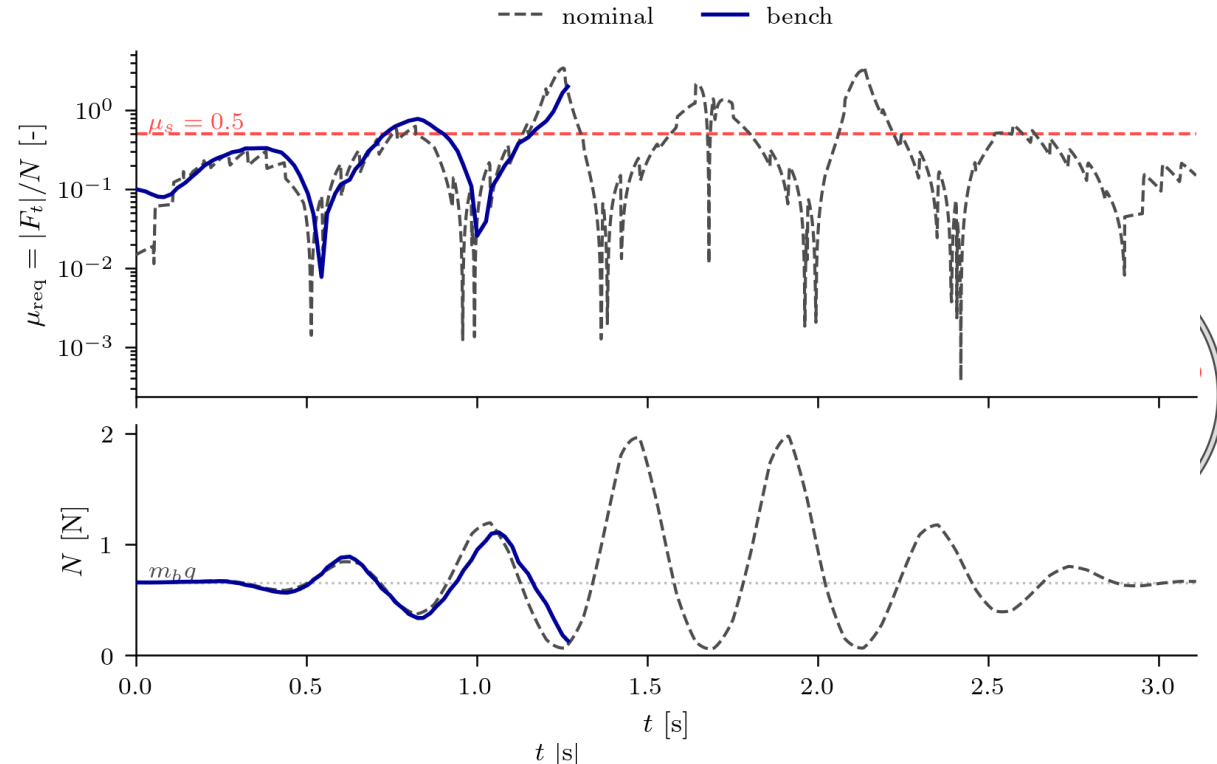
Continuous model error?

No major divergence up to failure.

Rolling constraint?

Friction coefficient μ exceeds literature value.

➔ MAYBE



Trajectories mathematically feasible, but physically impossible



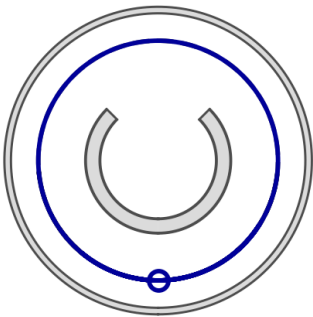
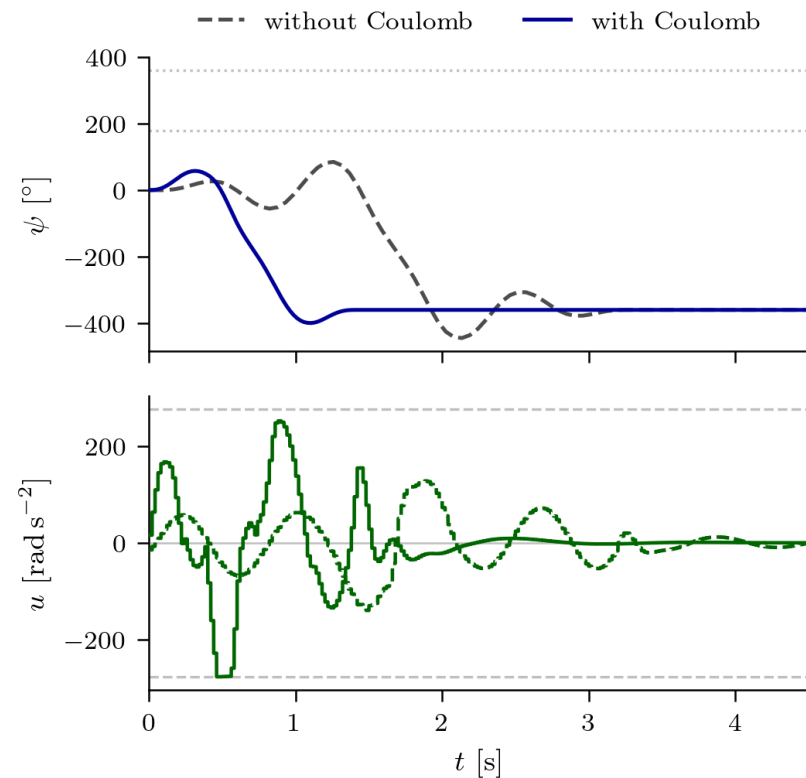
Looping Control Design

Planner restriction

Constraints addition

$$|F_t| \leq \mu N \text{ at every node and midpoint}$$

$$N \geq \varepsilon m_b g \text{ raised, } \varepsilon \geq 0$$



$$\mu_{req,max} = 3.7$$

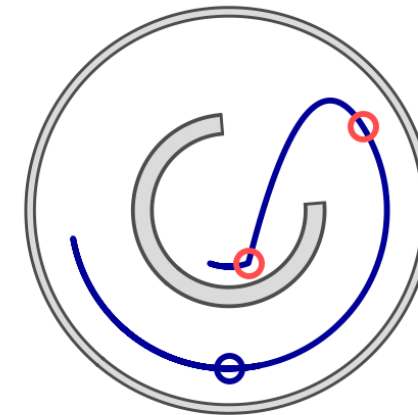
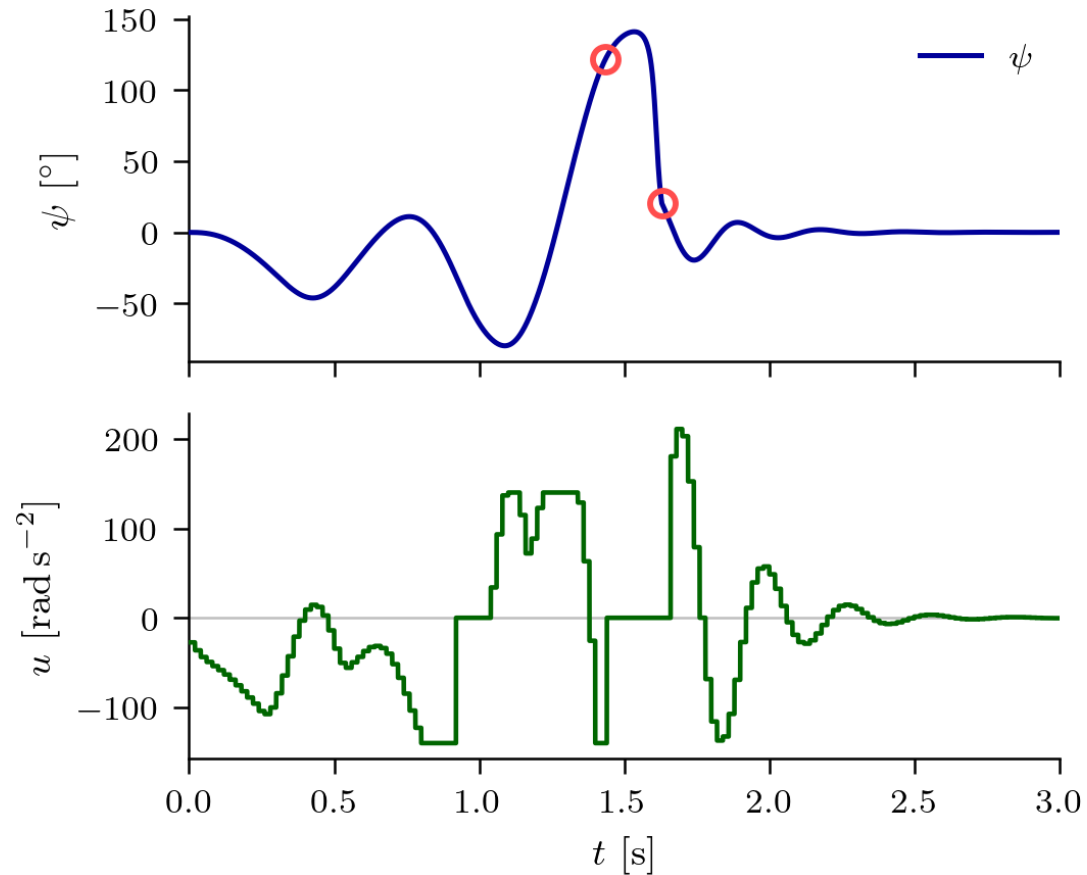
↓

$$\mu_{req,max} = 0.5$$

Value $\mu = 0,5$ taken from literature. Measuring it could also help investigate.

Flying Ball

Multi-phase problem



Problem solved and consistent, but inherits limitations from T3



Conclusion

What is needed to create a ball-and-hoop system?

What holds

- Hybrid model of four modes, verified numerically
- Bench built, modelled, and validated on the features tested
- All tasks succeed on simulation

Experimental Status

- T1 and T2 implemented on the bench
- T3 blocking is identified

Perspectives

- Address the identified friction limitation
- Extend the validation to the harder tasks

Overall, the ball-in-hoop is intuitive to new students and extensible to complex tasks



Thank you

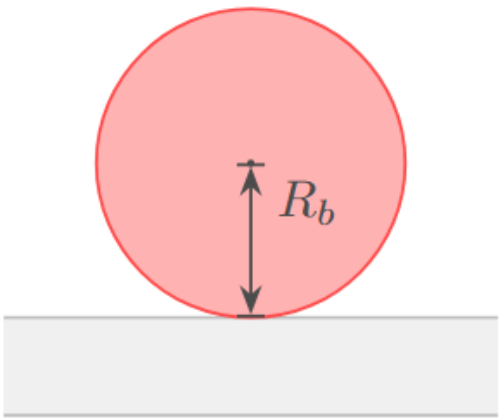
Do you have any question?

Prototype

Rails contact geometry

The contact is not a point

- O-rings tracks
- Ball must spin faster to travel the same distance
- Inertia is amplified (k)



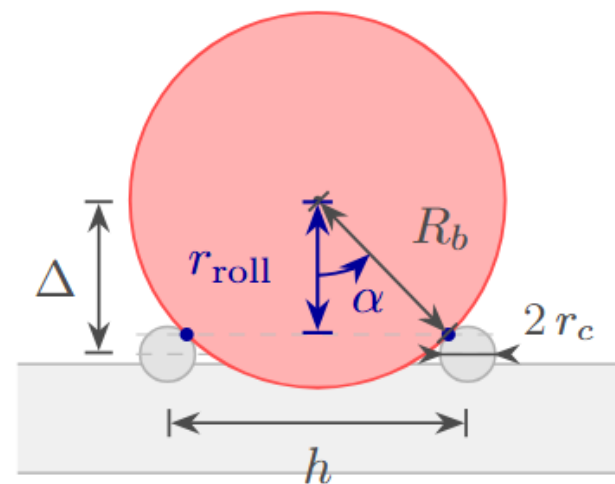
Ideal contact case

$$k = 1.4$$

$$f_0 = 1.24 \text{ Hz}$$



$$R_b > r_{roll}$$



Rails contact case

$$k = 1.552$$

$$f_0 = 1.32 \text{ Hz} \approx f_{0,bench} = 1.31 \text{ Hz}$$

$$k = 1 + \frac{2}{5} \left(\frac{R_b}{r_{roll}} \right)^2$$

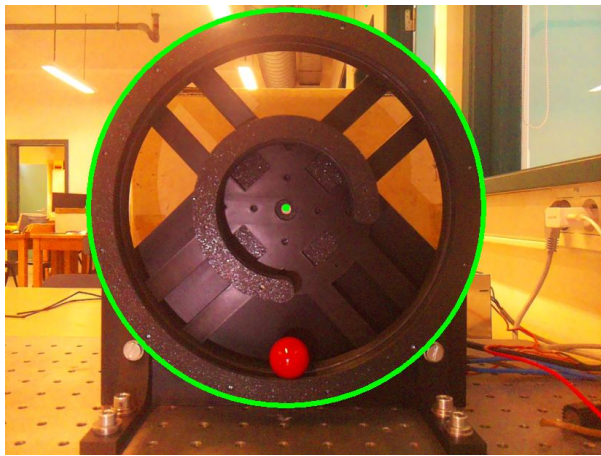
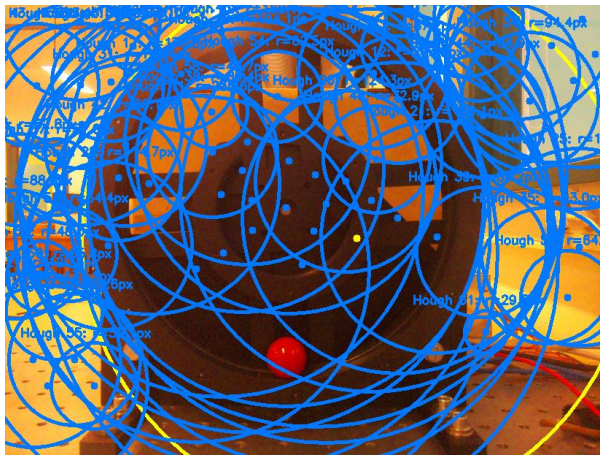
Prototype

Vision chain – centre, scale and segmentation

Centre and scale

- Rim detected by Hough transform
- Expected radius imposed, as unconstraining it leads to false positives
- Scale from the known hoop diameter

$$\lambda = 4.32 \cdot 10^{-4} \text{ m/px}$$

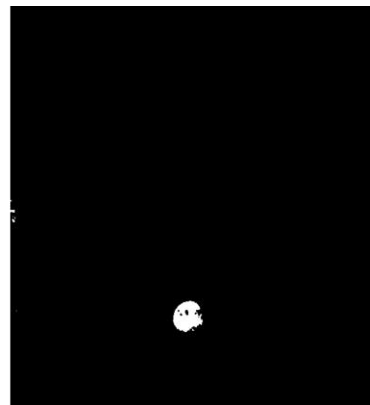
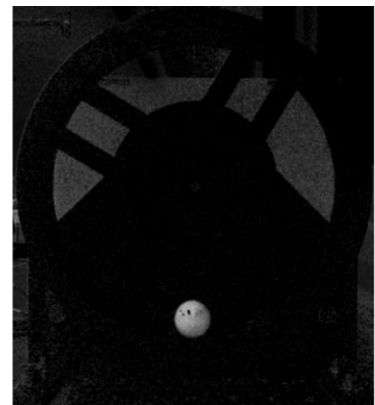


Repeatability

- **Centre:** 1.6 px RMS over 4 runs
→ 0.33° on ψ , against 1° target
- **Taubin fit:** 3.6 px apart

Segmentation

- Ball painted red, threshold 110
- Tracking window 64 px



ULB Prototype

E1 – Inertia and friction of the rotating assembly



Setup

- Hoops assembly and motor only
- Torque obtained from the driver current and constant

$$I_h \ddot{\theta} + b_m \dot{\theta} + \tau_c \text{sign}(\dot{\theta}) = \tau$$

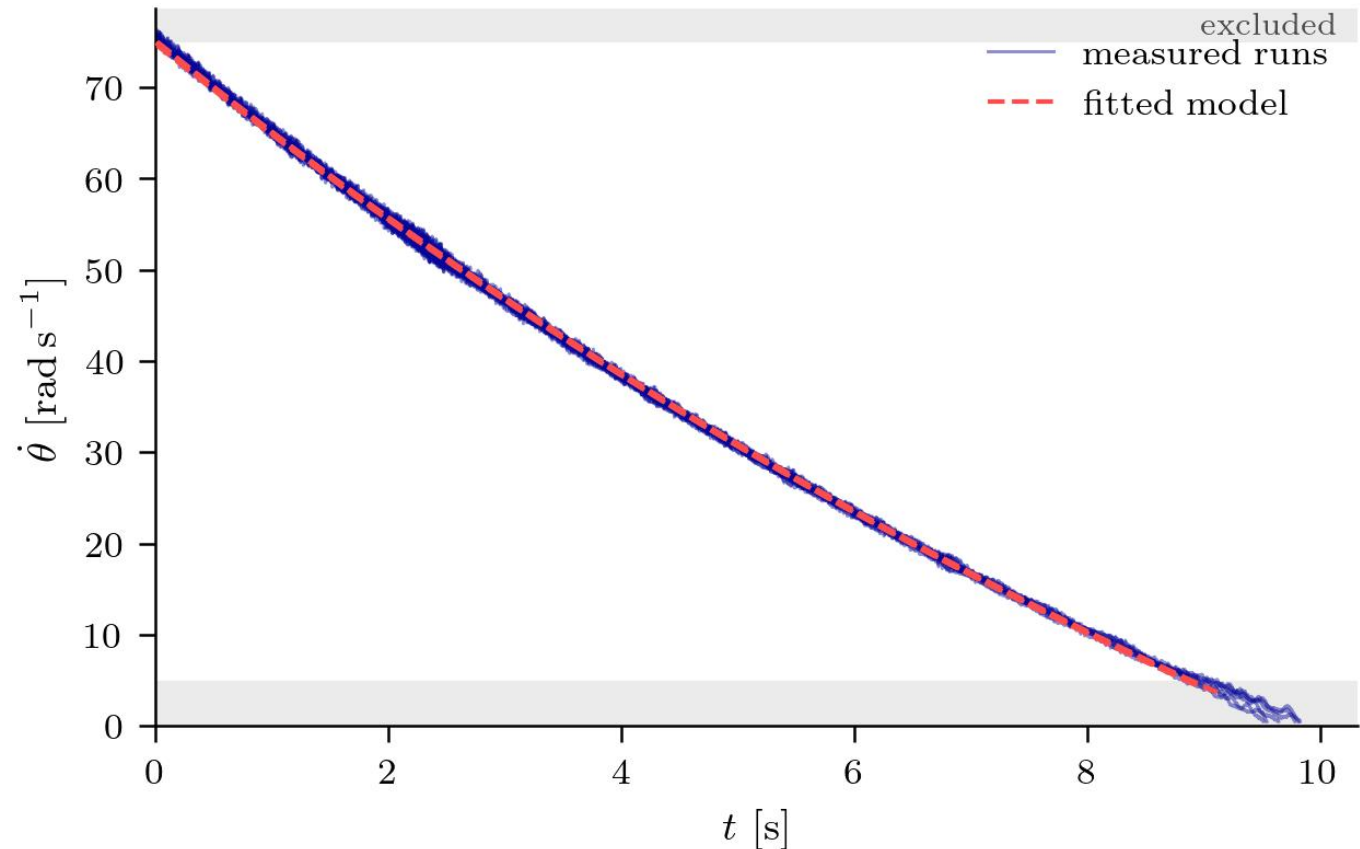
Steps

- **Steady State:** $I_h \ddot{\theta} = 0$,
 $b_m \dot{\theta} + \tau_c \text{sign}(\dot{\theta}) = \tau$

- **Free deceleration:**
 $I_h \ddot{\theta} + b_m \dot{\theta} + \tau_c \text{sign}(\dot{\theta}) = 0$

Results

$$\begin{aligned} I_h &= 1.12 \cdot 10^{-3} \text{ kg} \cdot \text{m}^2 \\ b_m &= 6.12 \cdot 10^{-5} \text{ N} \cdot \text{m} \cdot \text{s/rad} \\ \tau_c &= 5.38 \cdot 10^{-3} \text{ N} \cdot \text{m} \end{aligned}$$



Prototype

E2 – Friction of the ball

Setup

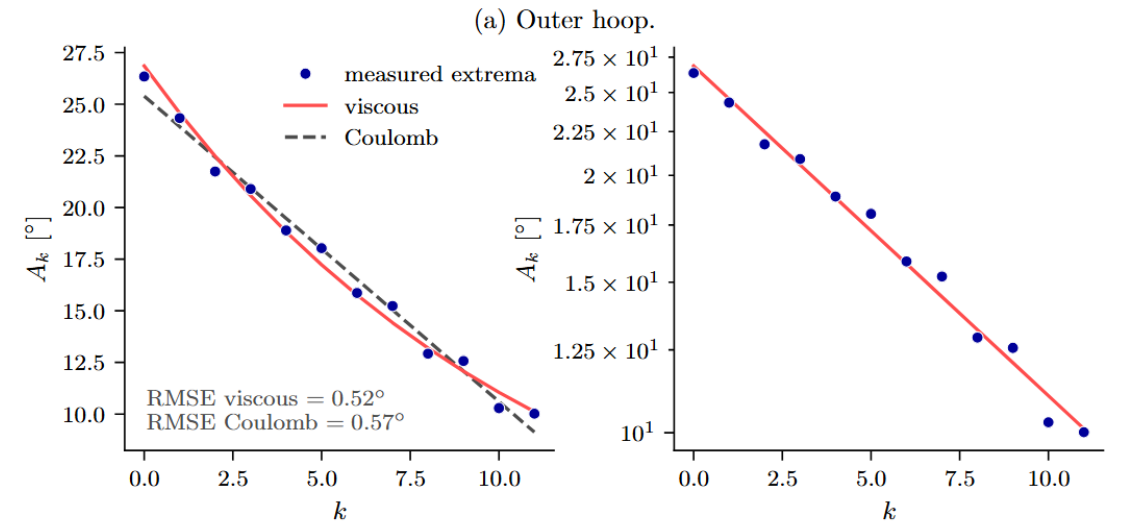
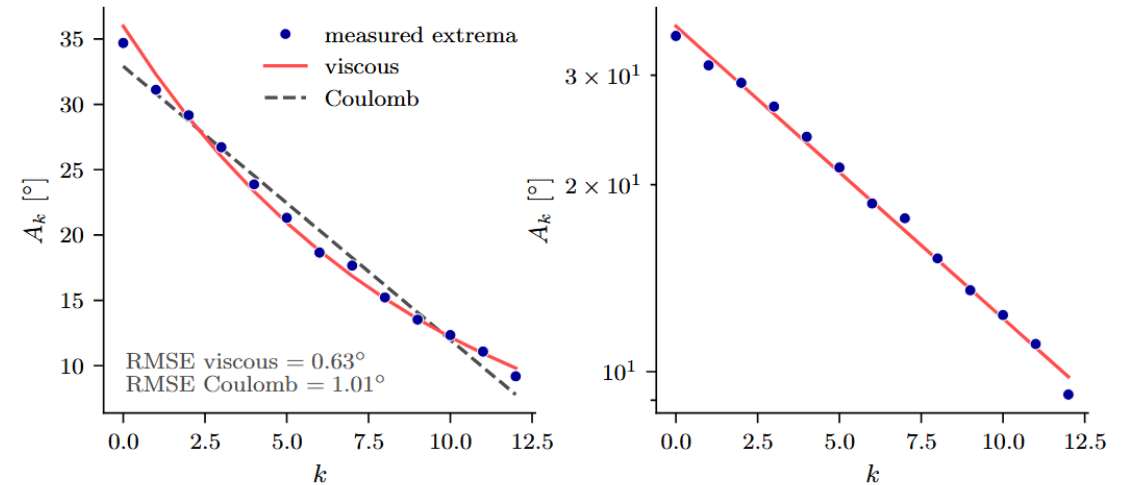
- Hoops blocked
- Ball present

Steps

- Ball released at rest, from $\psi_0 \approx 35^\circ$
- Free oscillations recorded
- Viscous $\rightarrow$ geometric decay
Coulomb $\rightarrow$ linear decay

Results

$$b_b = 6.48 \cdot 10^{-6} \text{ N} \cdot \text{m} \cdot \text{s/rad}$$



(b) Inner hoop.

Prototype

Parameters identification and validation

Identification

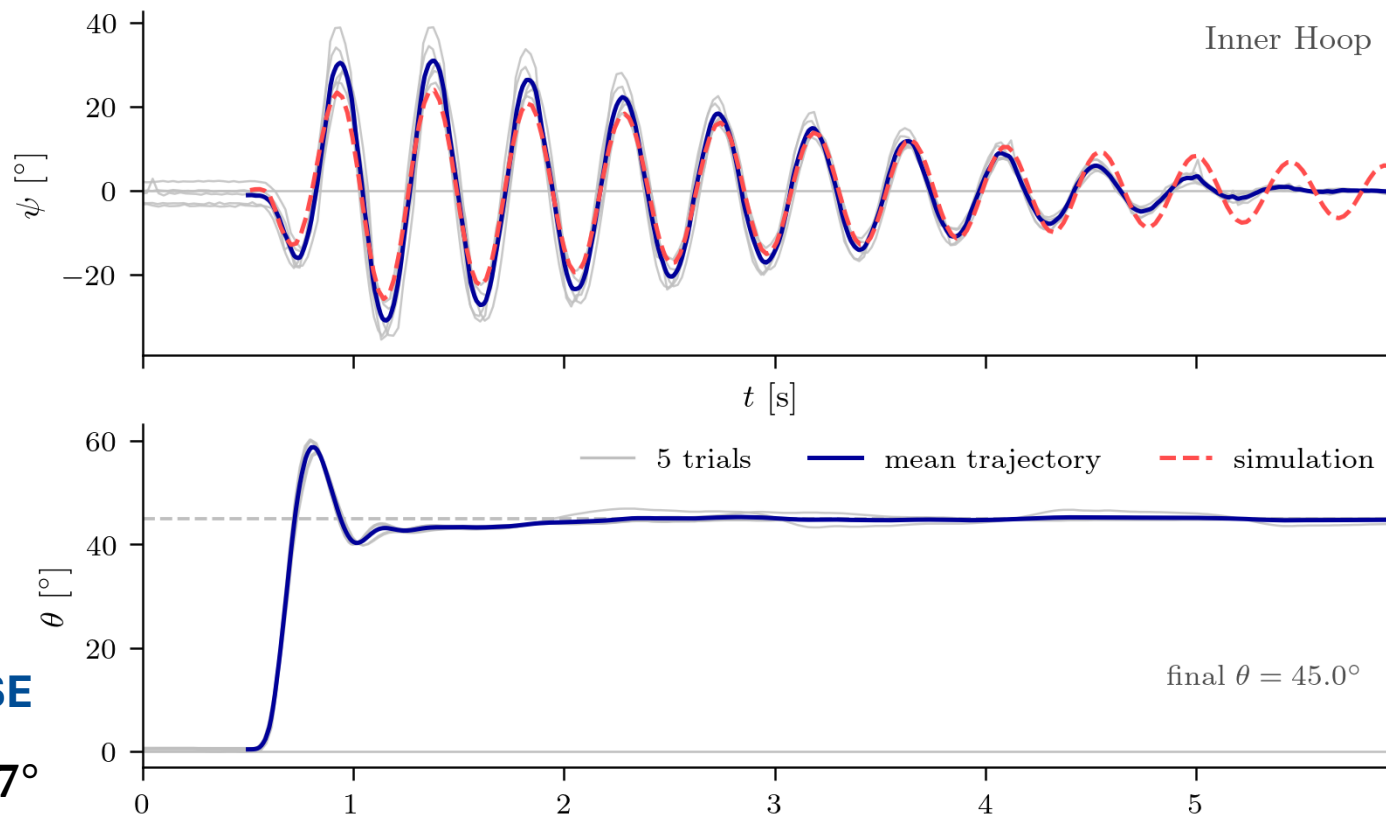
- Hoops and motor inertia
- Hoops and motor friction
- Ball friction

Protocol

- Simulation confronted with bench
- Criteria: $e(f_0) < 5\%$, $RMSE < 3^\circ$

Results

	$e(f_0)$	RMSE
Implemented	3.46%	2.47°
Specification	$\leq 5\%$	$\leq 5^\circ$



Linear Control Design

Linear Quadratic Regulation (LQR) open loop



Design loop

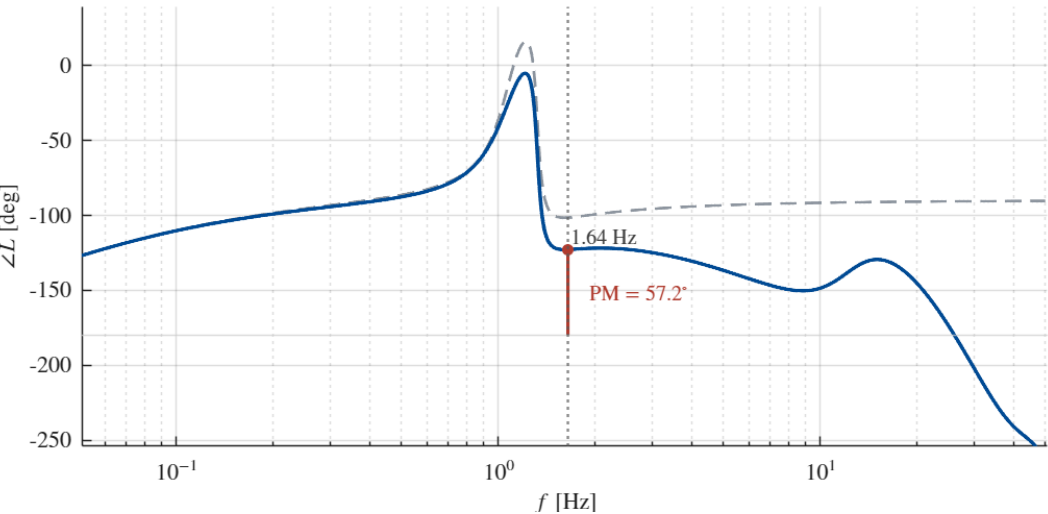
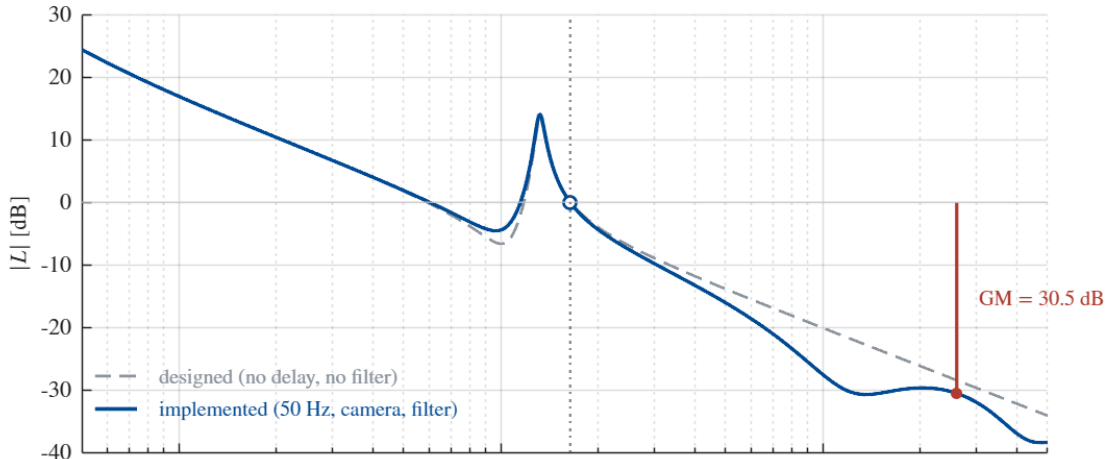
$$L(s) = K(sI - A)^{-1}B$$

Real loop

$$L(s) = \underbrace{K (sI - A)^{-1}B}_{\text{design}} \cdot \underbrace{e^{-j\omega\tau}}_{\text{cam}} \cdot \underbrace{e^{-\frac{j\omega T_s}{2}}}_{\text{ZOH}} \cdot \underbrace{\frac{1}{1 + j\omega/\omega_c}}_{\text{low-pass}}$$

Results

	GM	PM
Designed	∞	78.7°
Implemented	30.5 dB	57.2°
Specification	$\geq 6 \text{ dB}$	$\geq 45^\circ$



Control Design

Scale separation

Assumption

- Controller commands a hoop acceleration
- ➔ Driver must reach it before the command changes

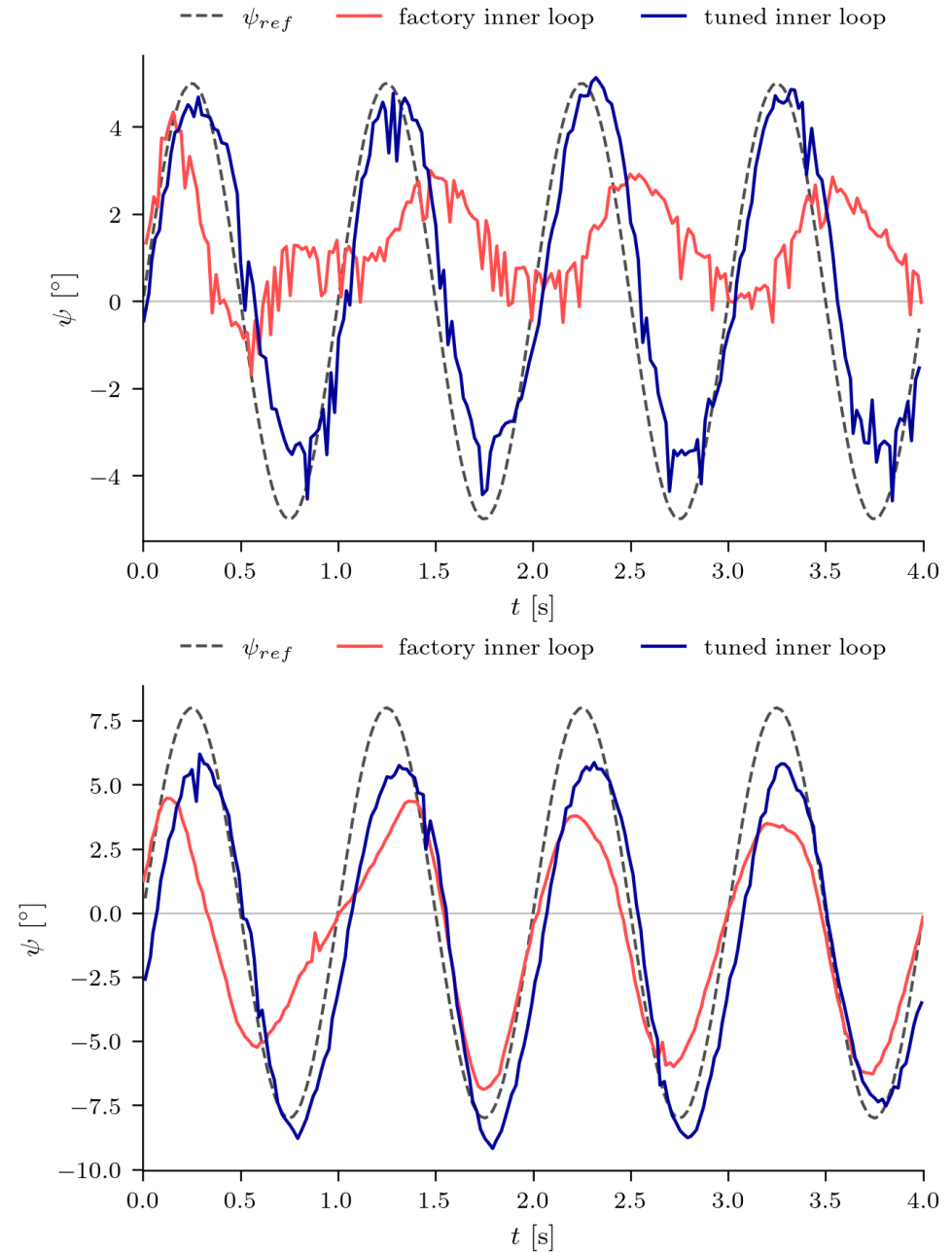
Measurements

	t_{inner}
Factory Gains	196 ms
Retuned Gains	17 ms

Impact

Same outer gains ➔ qualitatively different trackings

Nothing in the outer loop works if this inner loop is not fast enough



Linear Control Design

T1 - Overshoot and settling time

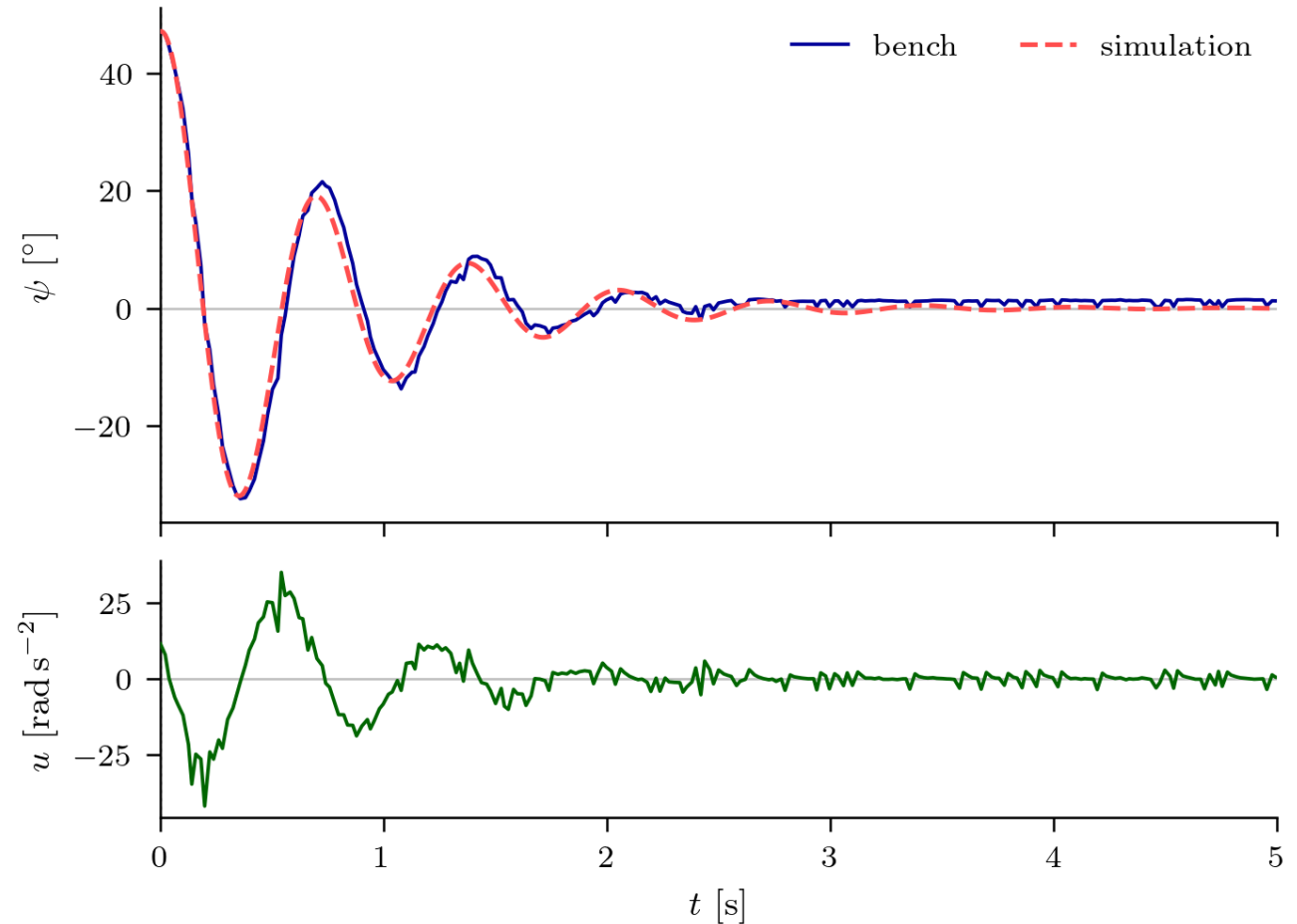


Results

	M_p	t_s
Designed	69.4 %	2.13 s
Implemented	84.3 %	2.18 s
Specification	$\leq 20 \%$	$\leq 3 \text{ s}$

Explanations

- **Design:** LQR weights favor t_s over M_p
- **Delay:** +15% at the bench



Looping Control Design

Expression of μ_{req}

Contributions

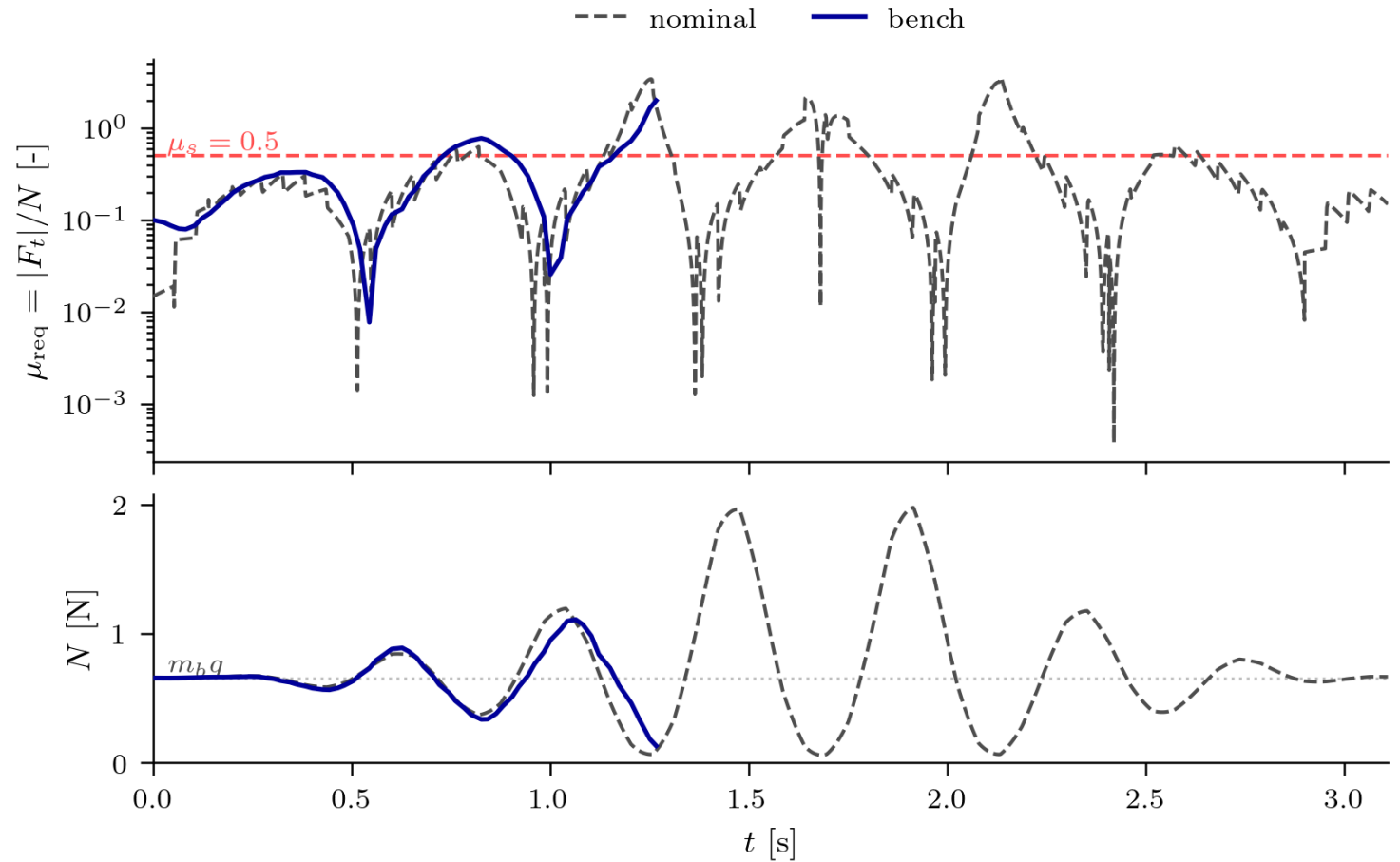
- Normal force: N
- Tangential force: F_t
- Gravity on the ball: $m_b g$

Total forces

$$\begin{cases} m_b R_{eff} \dot{\psi}^2 = N + m_b g \cos \psi \\ m_b R_{eff} \ddot{\psi} = F_t + m_b g \sin \psi \end{cases}$$

Friction coefficient

$$\mu_{req} = \frac{|F_t|}{N} = \frac{|R_{eff} \ddot{\psi} + g \sin \psi|}{g \cos \psi + R_{eff} \dot{\psi}^2}$$



Flying Ball

Multi-phase problem



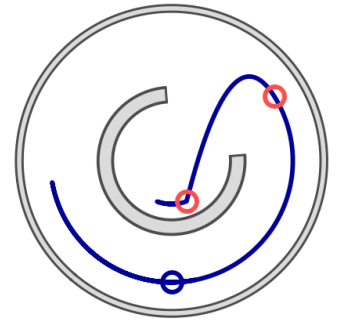
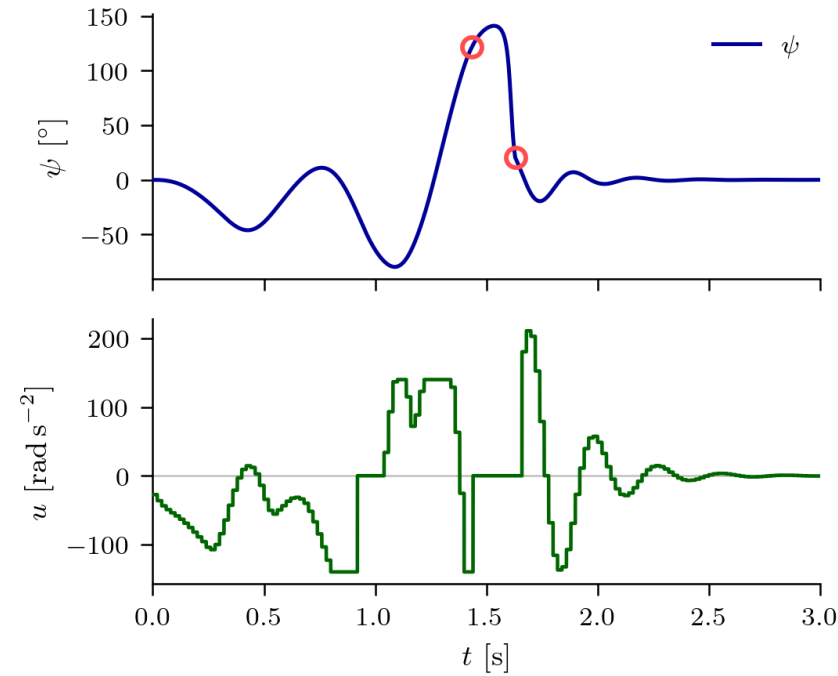
Problem formulation

State $x = (r, \dot{r}, \theta, \dot{\theta}, \psi, \dot{\psi})$, input $u = \ddot{\theta}$

$$\min_{x(\cdot), u(\cdot), T_{f,i}} J = \sum_{i=1}^3 \int_0^{T_{f,i}} u_i(t)^2 dt$$

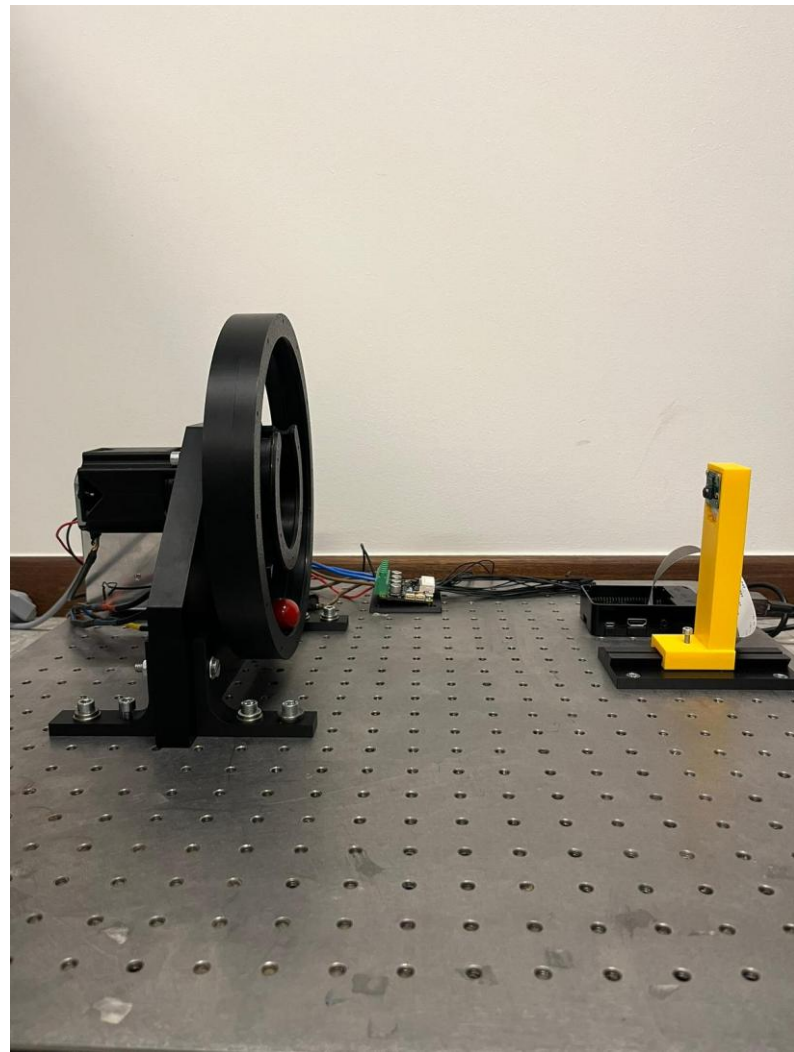
Under constraints

- Dynamics: $\dot{x} = f_i(x, u)$
- Actuation: $|u| \leq u_{max}$
- Boundaries: $x(0) = 0$, **outer hoop**
 $x(T_{f,3}) = 0$, **inner hoop, in**
- **Linking constraints (liftoff, impact)**



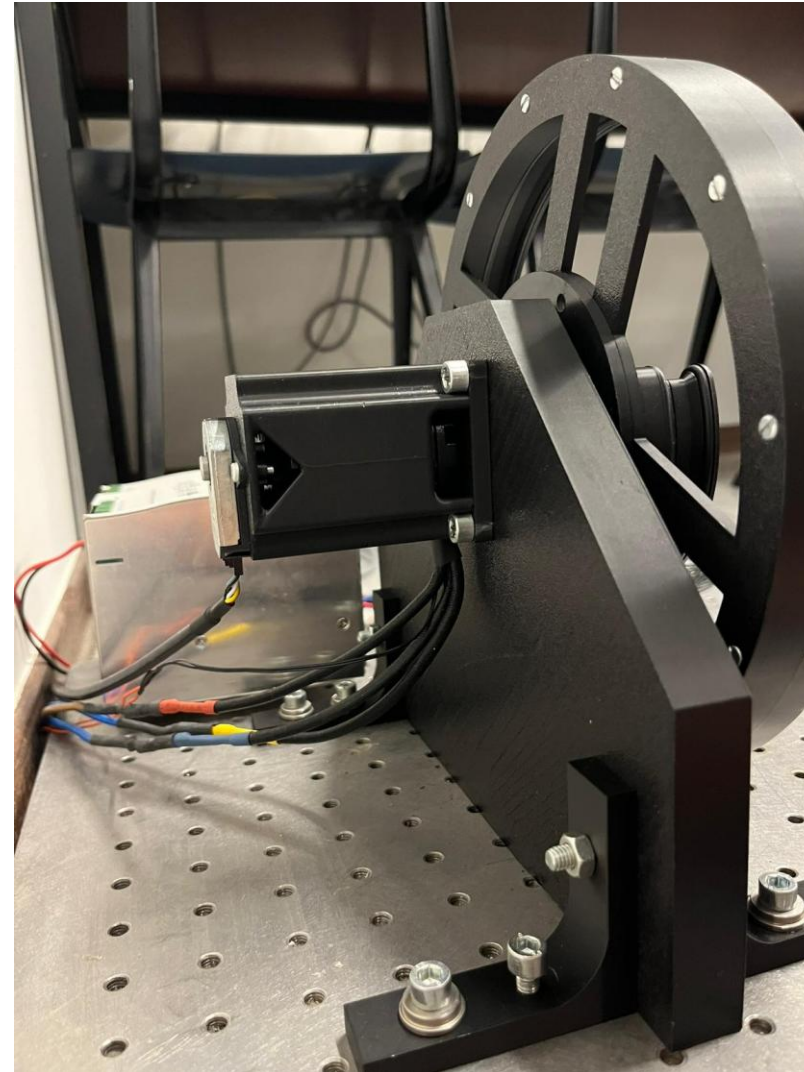
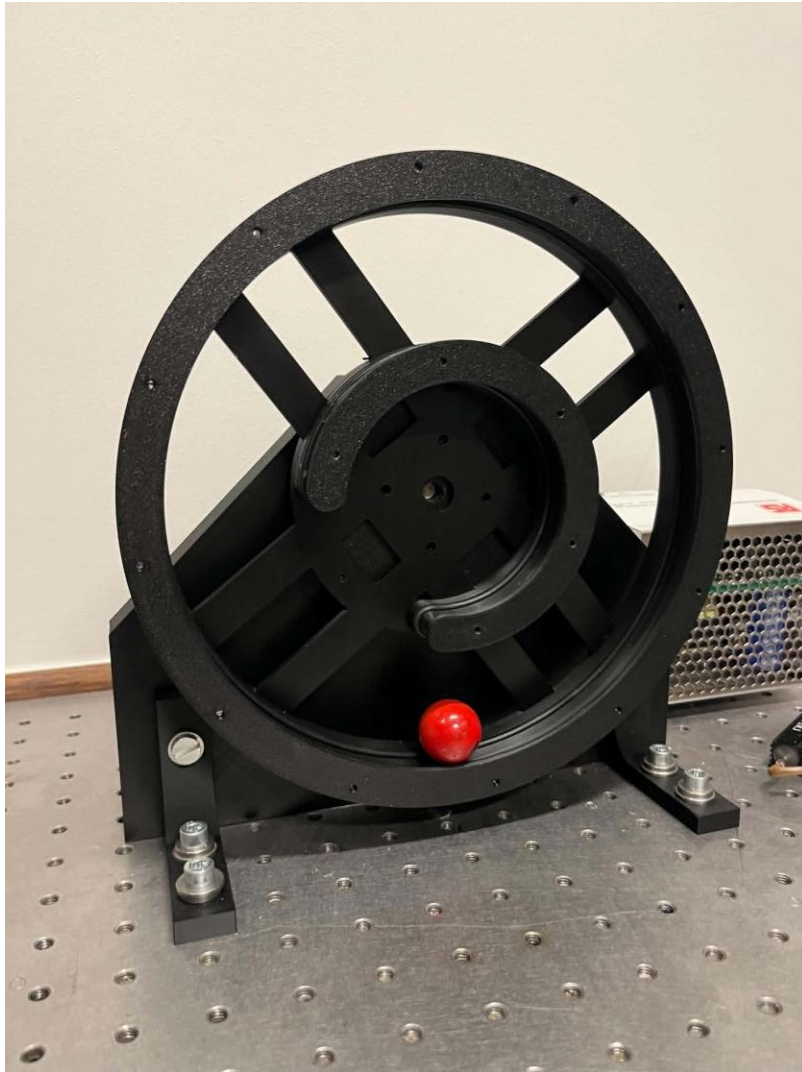
Prototype

Pictures of the Bench



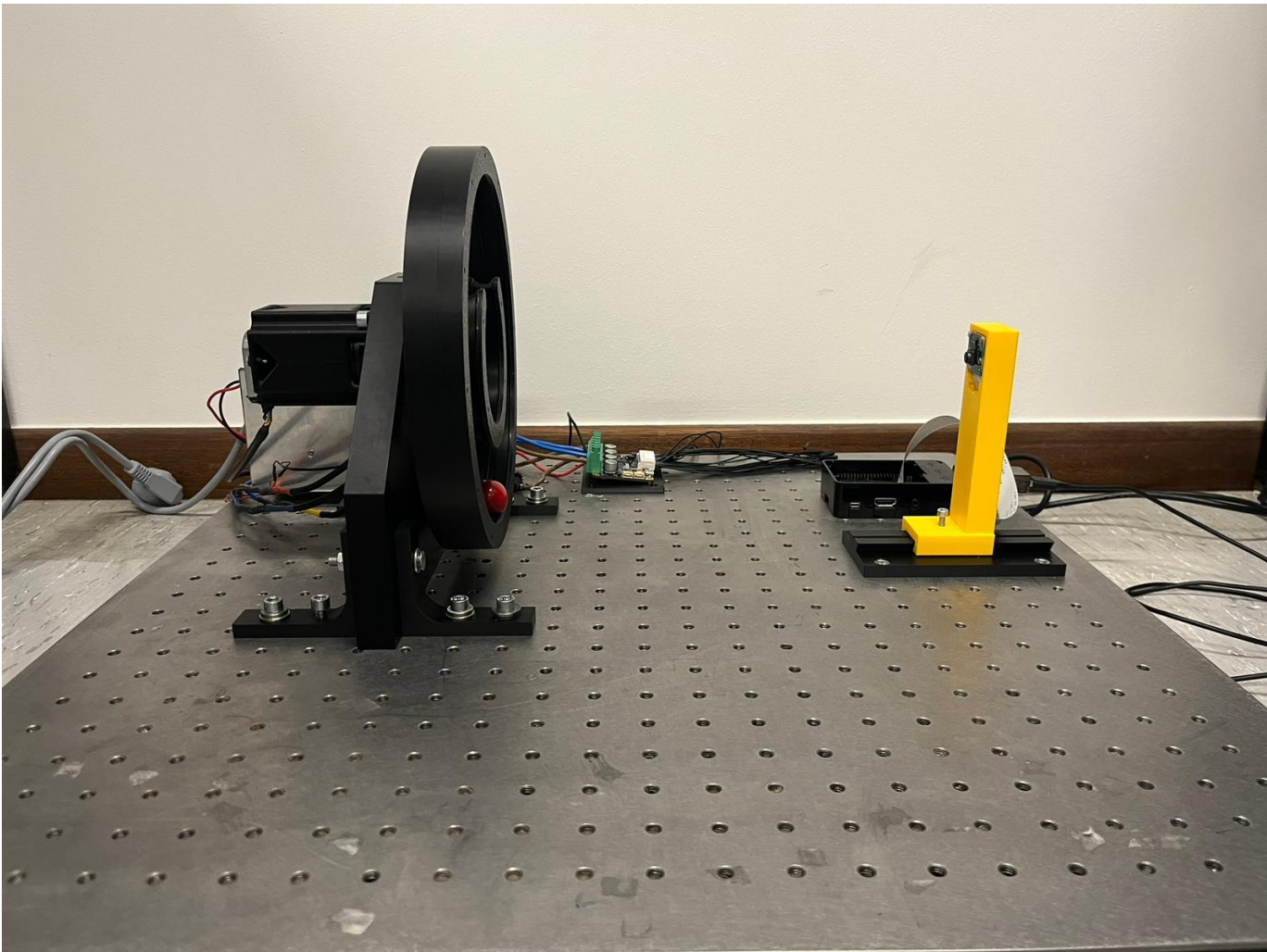
Prototype

Pictures of the Bench



Prototype

Pictures of the Bench



Control

Task 1 – Outer hoop



Free oscillations



Release from 10°

Control

Task 1 – Outer hoop



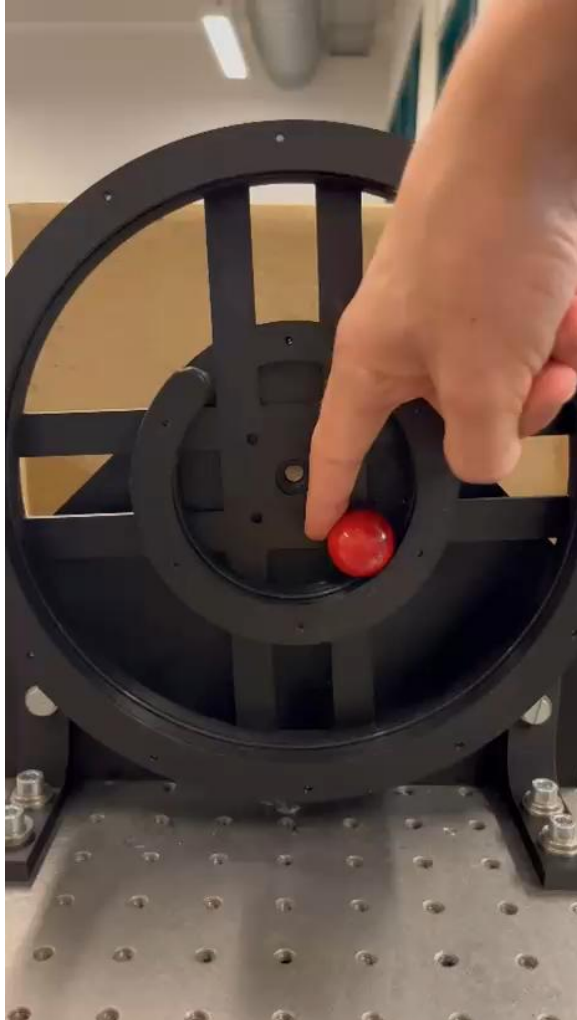
Release from 45°



Release from 60°

Control

Task 1 – Inner hoop



Release from 45°



Release from 60°

Control

Task 2



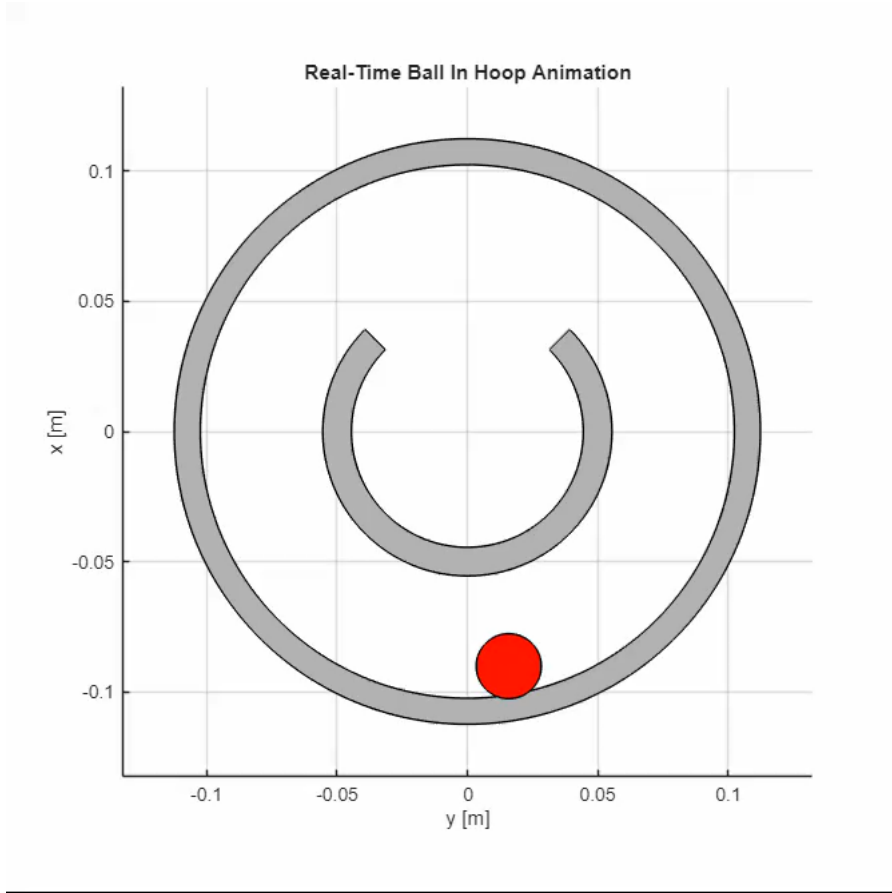
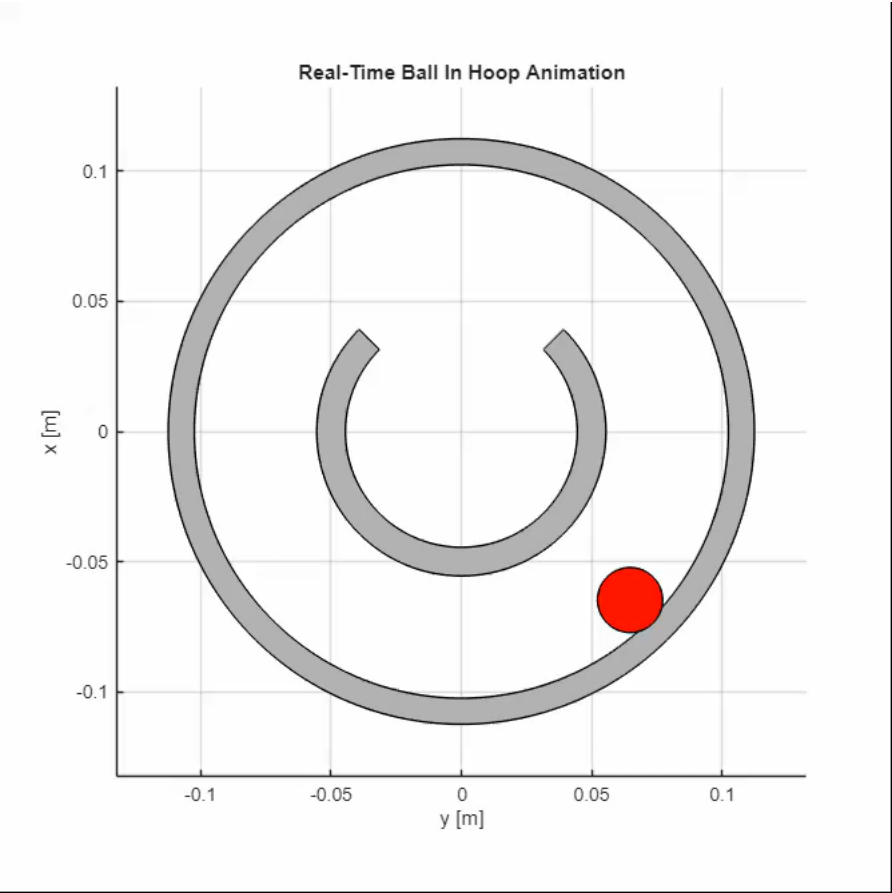
Inner hoop, 8° amplitude



Outer hoop, 8° amplitude

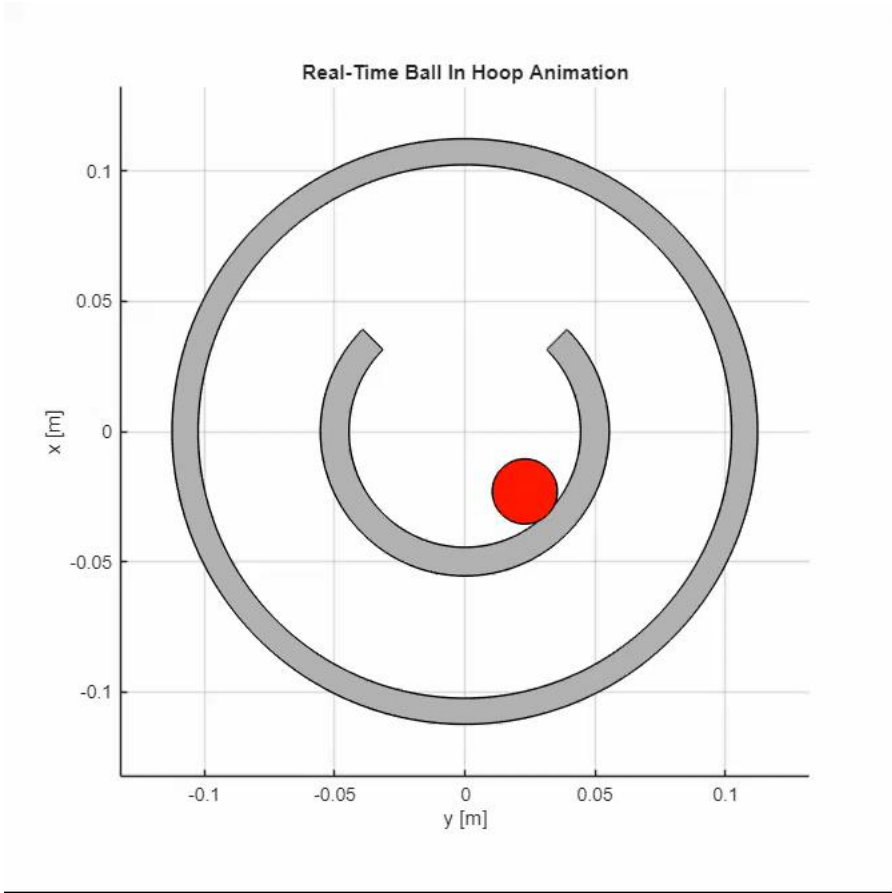
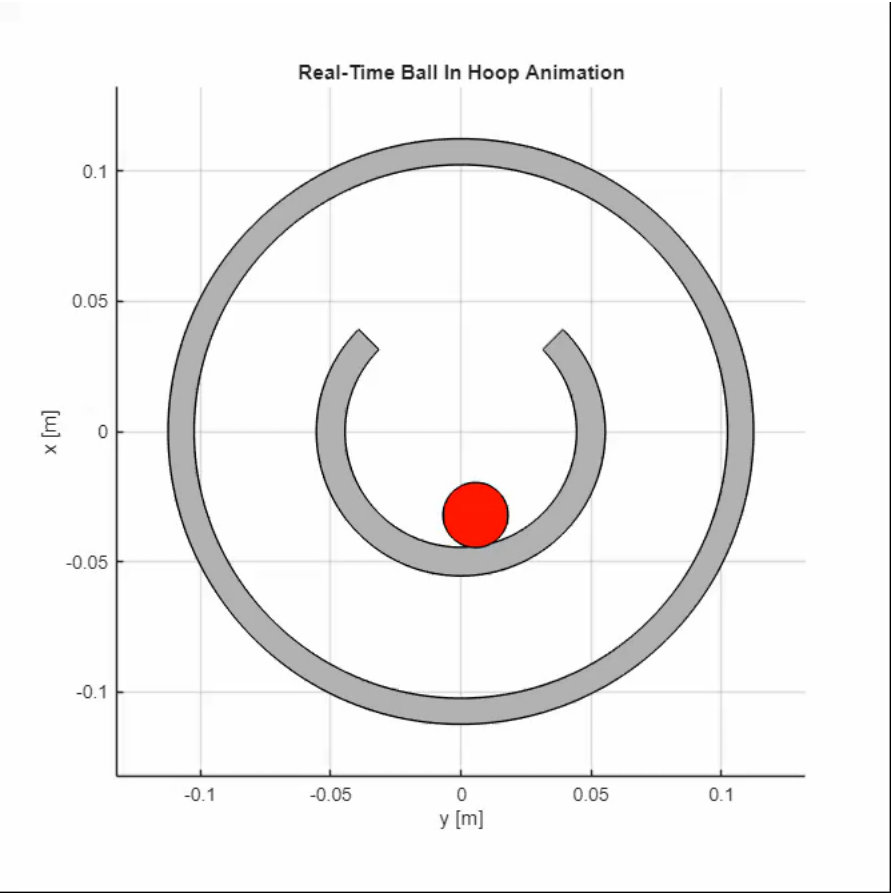
Simulation

T1 – Stabilisation – Outer hoop



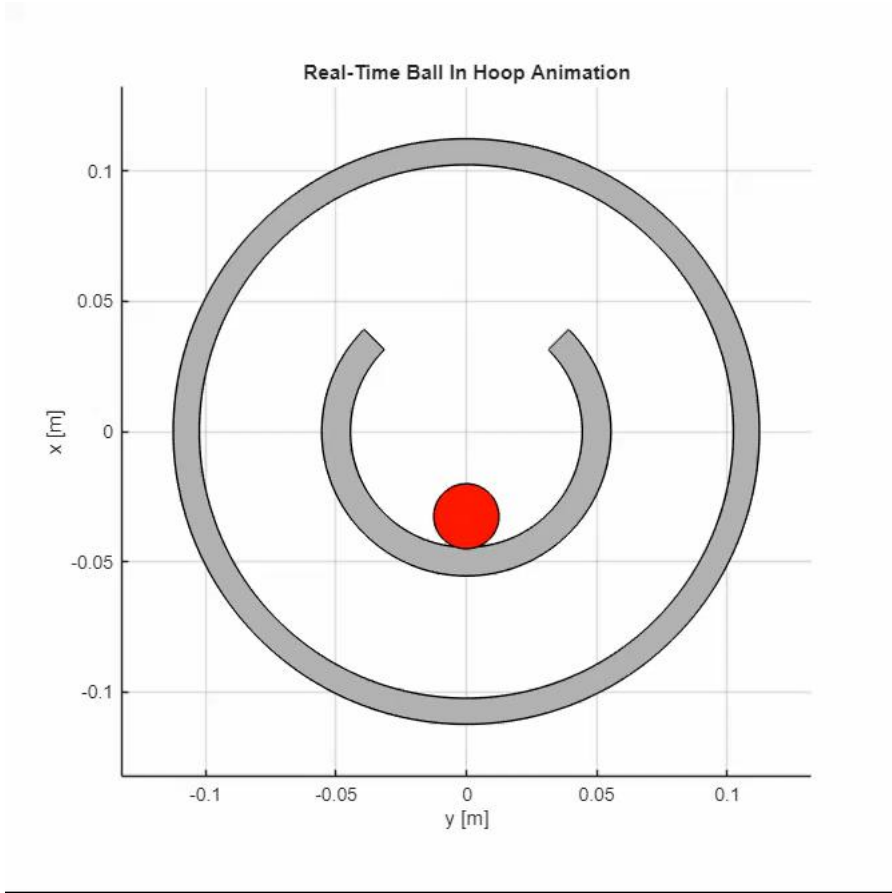
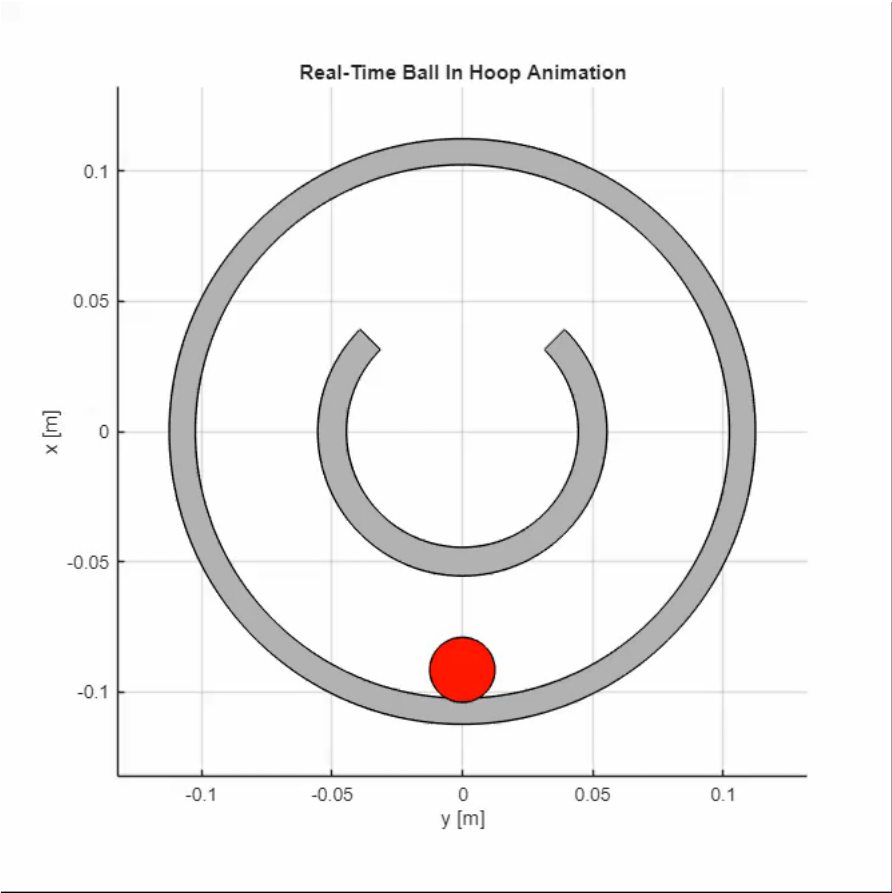
Simulation

T1 – Stabilisation – Inner hoop



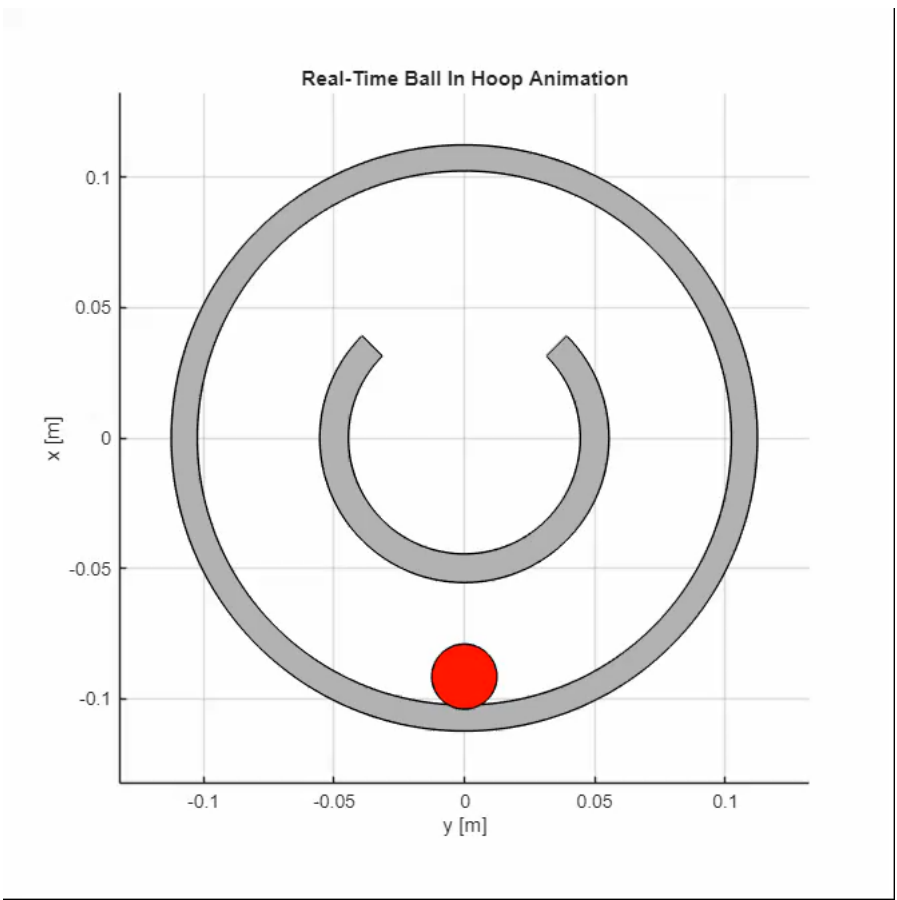
Simulation

T2 – Tracking around 0°



Simulation

T3 – Looping



Simulation

T4 – Flying Ball

